REPORT DOCUMENTATION PAGE

Form Approved
OMB No. 0704-0188

The public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Department of Defense, Washington Headquarters Services, Directorate for information on Operations and Reports (0704-0188), 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.
PLEASE DO NOT RETURN YOUR FORM TO THE ABOVE ADDRESS.

1. REPORT DATE *(DD-MM-YYYY)*	2. REPORT TYPE	3. DATES COVERED *(From - To)*
23-02-2022	Final	

4. TITLE AND SUBTITLE
Test Operations Procedure (TOP)
01-2-500B Transportability

5a. CONTRACT NUMBER

5b. GRANT NUMBER

5c. PROGRAM ELEMENT NUMBER

6. AUTHORS

5d. PROJECT NUMBER

5e. TASK NUMBER

5f. WORK UNIT NUMBER

7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES)
Aviation & Foreign Systems Division (TEAT-WFF) US Army Aberdeen Test Center
6943 Colleran Rd
Building 400
Aberdeen Proving Ground, MD 21005-5059

8. PERFORMING ORGANIZATION REPORT NUMBER
TOP 01-2-500B

9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)
Policy and Standardization Division (CSTE-CI-P)
U.S. Army Test and Evaluation Command
6617 Aberdeen Boulevard
Aberdeen Proving Ground, MD 21005-5001

10. SPONSOR/MONITOR'S ACRONYM(S)

11. SPONSOR/MONITOR'S REPORT NUMBER(S)
Same as item 8

12. DISTRIBUTION/AVAILABILITY STATEMENT

Distribution Statement A. Approved for public release; distribution is unlimited.

13. SUPPLEMENTARY NOTES
Defense Technical Information Center (DTIC), AD No.: AD1161671

This TOP supersedes TOP 01-2-500A, dated 23 April 2013.

Marginal notations are not used in this revision to identify changes, with respect to the previous issue, due to the extent of the changes.

14. ABSTRACT
This TOP provides guidance for preparing test plans and conducting test programs to evaluate the transportability characteristics of military equipment whether towed, self-propelled, or moved by carrier over highway, off-road terrain, railway, waterway, or by air.

15. SUBJECT TERMS
fixed wing; rotary wing; air dropped containers; shelter free fall; rail movement; lift provisions; tie-down provisions

16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES	19a. NAME OF RESPONSIBLE PERSON
a. REPORT	B. ABSTRACT	C. THIS PAGE			19b. TELEPHONE NUMBER *(include area code)*
Unclassified	Unclassified	Unclassified	SAR	77	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 01-2-500B
DTIC AD No. AD1161671

23 February 2022

TRANSPORTABILITY

Paragraph			Page
	1.	SCOPE	2
	2.	FACILITIES AND INSTRUMENTATION	3
	2.1	Facilities	3
	2.2	Instrumentation	7
	3.	REQUIRED TEST CONDITIONS	11
	4.	TEST PROCEDURES	13
	4.1	Lifting and Tie-down Provisions	13
	4.2	Rail Transportability	14
	4.3	Highway Transportability	18
	4.4	Marine Transportability	18
	4.5	Terminals Handling and Movement	20
	4.6	Air Transportability - Fixed Wing Internal	22
	4.7	Air Transportability - Rotary Wing/Tiltrotor Internal	26
	4.8	Air Transportability - Rotary Wing/Tiltrotor External	27
	4.9	Air Transportability - Airdropped Materiel	28
	5.	DATA REQUIRED	30
	5.1	Lifting and Tie-down Provisions	30
	5.2	Rail Transportability	30
	5.3	Highway Transportability	31
	5.4	Marine Transportability	31
	5.5	Terminals Handling and Movement	32
	5.6	Air Transportability - Fixed Wing Internal	33
	5.7	Air Transportability - Rotary Wing/Tiltrotor Internal	33
	5.8	Air Transportability - Rotary Wing/Tiltrotor External	34
	5.9	Air Transportability - Airdropped Materiel	34
	6.	PRESENTATION OF DATA	35
	6.1	Lifting and Tie-down Provisions	35
	6.2	Rail Transportability	38
	6.3	Highway Transportability	41
	6.4	Marine Transportability	41
	6.5	Terminals Handling and Movement	41
	6.6	Air Transportability - Fixed Wing Internal	41
	6.7	Air Transportability - Rotary Wing/Tiltrotor Internal	42

*This TOP supersedes TOP 01-2-500A, dated 23 April 2013.

Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

			Page
Paragraph	6.8	Air Transportability - Rotary Wing/Tiltrotor External	42
	6.9	Air Transportability - Airdropped Materiel	45
APPENDIX	A.	GENERAL PHYSICAL CHARACTERISTICS FOR TRANSPORTABILITY	A-1
	B.	FIXED WING AIR TRANSPORTABILITY REQUIREMENTS	B-1
	C.	ROTARY-WING AIRCRAFT INTERNAL PAYLOAD CAPABILITIES	C-1
	D.	ROTARY-WING AIRCRAFT EXTERNAL PAYLOAD CAPABILITIES	D-1
	E.	EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA	E-1
	F.	ABBREVIATIONS	F-1
	G.	REFERENCES	G-1
	H.	APPROVAL AUTHORITY	H-1

1. SCOPE.

a. This Test Operations Procedure (TOP) provides guidance for evaluating the transportability characteristics of military equipment whether towed, self-propelled, or moved by carrier over highway, off-road terrain, railway, waterway, or by air. Transportability is an important factor in maintaining the "inventory in motion" concept of the modern, highly mobile army. Materiel must be able to survive transportation in a military environment without a reduction in functional performance. The environment imposes numerous constraints involving impacts, vibrations, interferences, and repetitive motions requiring attention to blocking, bracing, lifting, tie-down, and containerization in connection with the stowage, orientation, suspension, or transfer of cargo. Adequacy of design depends upon compatibility with the transportation media and performance after handling and transport.

b. All tests specified herein are not applicable to all test items. The test planner will be selective to include only those tests needed to satisfy the requirements for the specific item to be tested. Data from previous and similar tests and data obtained by concurrent testing will be considered to avoid duplication and reduce the scope of testing.

c. The appropriate Department of Defense (DOD) component mode transportability office will be consulted by materiel developers for transportability guidance and approval of their equipment that meets the definition of a transportability problem item as outlined in Department of Defense Instruction (DODI) 4540.07[1]**.

** Superscript numbers correspond to those in Appendix G, References.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

d. Testing will provide each DOD component mode transportability office the data necessary to analyze and assess the transportability capabilities and limitations of the equipment. The agencies and their responsibilities are referenced in applicable test procedures herein. Direct consultation with each office for guidance and verification of requirements and procedures is encouraged.

e. Military Standard (MIL-STD)-1366E[2], Transportability Criteria, is the primary document for all modes of transport per DODI 4540.07. This standard establishes basic transportability interface criteria for use by the DOD acquisition community, to include the private sector, in the development and shipment of items of materiel. The standard covers dimensional and weight limitations for all modes of transport to ensure that new and modified systems meet the interface requirements of the Defense Transportation System (DTS) and the DTS lift assets. It also covers lifting and tie-down provisions, containerization criteria, overloads, assembly/disassembly, air delivery, shelter criteria, transportability testing, and modeling and simulation of the transportation environment.

f. In regard to Terminals Handling and Movement, the International Organization for Standardization (ISO) Standard 1496/1:2013[3] contains specifications and tests that are required for American National Standards Institute/International Organization for Standardization (ANSI/ISO) certification of all intermodal freight containers, but the contents may require rail impact testing. The Military Surface Deployment and Distribution Command, Transportation Engineering Agency (SDDCTEA) accepts such certification of intermodal freight containers in lieu of additional transportability testing. Therefore, the provisions of this TOP need not be applied to certified containers, with exception to the internal contents of the container. In the event certification is sought for a container or rigid-wall shelter, a list of ANSI/ISO test facilities may be obtained through SDDCTEA, ATTN: SDTE-DPE, 1 Soldier Way, Building 1900W, Scott AFB, IL 62225-5006. Additionally, shelter transportability certifications must be obtained from the U.S. Army Combat Capabilities Development Command (DEVCOM) Soldier Center (SC), Natick, MA.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

2.1.1 Lifting and Tie-down Provisions.

Item	Requirement
Bedplate/embedded channels.	Provide a means of mounting equipment for vertical restraint.
Deadman anchors/upright restraints.	Provide a means of mounting equipment for longitudinal and lateral pull restraints.
Boom/overhead crane.	Provide a means for vertical load application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Item	Requirement
Horizontal load application device.	Provide a means for longitudinal/ lateral load application.
Restraint rigging (i.e., chains, straps).	Secure the test item to the bedplate or deadman anchor for vertical, longitudinal, or lateral load restraint.
Lift rigging (i.e., lifting chain or sling).	Secure the test item to the proper mounting equipment and load application device for vertical loading.
Video monitoring equipment.	Provide the ability to view testing for real-time feedback.

2.1.2 Rail Transportability.

Item	Requirement
Straight, dry, level section of railroad track 61 meters (m) (200 feet (ft)) minimum length, with additional track in order to reverse the test railcar/buffer car(s) for impacting in the opposite direction.	Provide distance to accelerate locomotive and the test item railcar to specified impact speeds. Ensure railroad system complies with criteria per Unified Facilities Criteria (UFC) 4-860-03[4] for safe test operations.
Mechanical drive (i.e., locomotive, tug, inclined ramp).	Accelerate the railcar with the test item to the required pre-impact velocity.
Test railcar equipped with end-of-car cushioned draft gear and a conventional underframe (without cushioning) and chain tie-downs.	Transport the test item into the buffer car(s).
Buffer car(s) with minimum weight of 113,400 kilograms (kg) (250,000 pounds (lb)), with the first buffer car equipped with standard (friction) draft gear and remaining (up to 5) buffer cars (if any) equipped with either standard draft gear or cushioned draft gear, with the draft gear compressed for either case.	Act as buffer car(s) into which the test item car is impacted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Item	Requirement
Temperature conditioning chamber (if required).	Provide the required temperature environment for the test item.
Video and photographic recording equipment.	Provide video and photographic documentation for review and analysis.

2.1.3 Highway Transportability.

Item	Requirement
Automotive road courses.	Conduct road transport and mobility/ handling testing.
Recovery vehicles.	Conduct recovery vehicle test operations.

2.1.4 Maritime Transportability.

Item	Requirement
Maritime vessel.	Provide representative vessel for loading characteristics.
Crane/lifting device (with associated rigging).	Provide means to maneuver test item onto/ within vessel.
Dockside or on-board activity environment.	Adequate loading/ unloading zone for test activity.

2.1.5 Terminals Handling and Movement.

Item	Requirement
Material handling equipment (MHE).	Transportation of item(s).
Concrete pad or adequate test pad.	Operations and/or lifting and drop test location.
Rigging equipment (sling, cables, chain, fittings).	Lifting operation and drop test.
Quick-release hook (or similar device).	Mechanism for release of item in support of drop testing.
Towing bar bridle and safety chains.	Skid resistance testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

2.1.6 Air Transportability - Fixed Wing Internal.

Item	Requirement
Fixed-wing aircraft.	Provide representative aircraft for loading exercise.
Level loading surface.	Provide representative loading facility/ environment for loading activity.
Cargo handling equipment.	Assist in loading material in/out of aircraft.

2.1.7 Air Transportability - Rotary Wing Internal.

Item	Requirement
Military cargo helicopter.	Perform form-fit function test loading with functioning loading facilities.
Level loading surface.	Provide representative loading facility/environment for loading activity.
Cargo handling equipment.	Assist in loading material in/out of aircraft.
Tie-down fittings and tie-down devices.	Restrain item to helicopter's cargo deck.

2.1.8 Air Transportability - Rotary Wing External.

Item	Requirement
Military cargo helicopter.	Perform flight demonstration.
Controlled airspace.	Serve as designated flight area.
Helipad or airstrip.	Serve as designated staging, pickup, and delivery area.
Chase aircraft (with video equipment).	Documentation of flight (video and photographic).
Boom/overhead crane.	Provide lifting capability for static configuration trials.
Spreader bar or theodolite.	Simulate helicopter's cargo hook spacing for dual-point lift configuration.
Standard military sling sets.	Rigging of test item for static/ flight demonstration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Item	Requirement
Rigging materials including cargo straps, nylon cord, cotton webbing, and tape.	Restraint of all internal and external components for flight test.
Video recording equipment.	Video documentation (ground/air perspectives) for review and analysis.

2.1.9 Air Transportability - Airdropped Materiel.

Item	Requirement
Controlled airspace and ranges	Serve as designated flight area and drop zones.
Fixed-wing aircraft	Perform airdrop/low altitude parachute extraction (LAPE) or low velocity air drop (LVAD) activities.
Static drop facility.	Perform ground impact test.
Slings, parachutes, and/or airdrop platforms	Rigging of test item for airdrop.
Rigging materials	Restraint and cushioning of all components for airdrop.
Boom/overhead crane.	Load placement for rigging.
Aircraft loader.	Load items into aircraft.
High-speed film/video equipment.	Document airdrop characteristics.

2.2 Instrumentation.

2.2.1 Lifting and Tie-down Provisions.

Devices for Measuring	Measurement Accuracy
Load cell (force)	± 1% of reading
Caliper (length)	± 0.025 centimeter (cm) (± 0.001 inch (in.))
Tape (length)	± 0.159 cm (± 0.0625 in.)
Inclinometer (angle)	± 0.1 degree

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Devices for Measuring	Measurement Accuracy
Timer (elapsed time)	± 1 second
Scale (weight)	± 0.5% of reading

2.2.2 Rail Transportability.

Devices for Measuring	Measurement Accuracy
Scale (weight)	± 0.5% of reading
Timer (rail impact speed)	± 0.16 kilometer per hour (km/hr) (± 0.1 mile per hour (mph))
Speedometer (locomotive speed)	± 0.16 km/hr (± 0.1 mph)
Strain-gauged coupler (force, as required)	±2.2 kilonewton (kN) (± 500.0 pound force (lbf))
Torque wrench (wire rope clamp, as required)	± 5% of reading
Accelerometer (acceleration, as required)	± 0.05 *g* (*g* is the acceleration due to gravity at the Earth's surface)
Strain gauges (strain, as required)	± 5 microstrain
Thermocouple (temperature, as required)	± 2.0 °Celsius (°C) (± 3.6 °Fahrenheit (°F))

2.2.3 Highway Transportability.

Devices for Measuring	Measurement Accuracy
Accelerometer (deceleration)	± 0.05 *g*
Fifth wheel (distance, brake stopping)	± 0.1 m (± 0.3 ft)
Walking wheel (distance, road)	± 0.1 km (± 0.06 mile)
Speedometer (speed)	± 0.1 km/hr (± 0.06 mph)
Scale (weight)	± 0.5% of reading
Tape (physical dimensions)	± 0.159 cm (± 0.06 in.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

2.2.4 Marine Transportability.

Devices for Measuring	Measurement Accuracy
Approach/departure/ breakover angles of test item	± 1 degree
Ground clearance of test item	± 1 cm (± 0.4 in.)
Time (rigging time)	± 1 minute
Time (voyage time)	± 1 minute
Acceleration/deceleration	± 1% of reading
Strain	± 1% of reading
Vessel pitch, roll and yaw angles	± 1 degree
Ramp angles of approach, departure and crest	± 1 degree

2.2.5 Terminals Handling and Movement.

Devices for Measuring	Measurement Accuracy
Accelerometer (acceleration)	± 0.05 *g*
Load cell (force)	± 1% of reading
Scale (weight)	± 1% of reading
Tape (height, drop test)	± 0.159 cm (± 0.06 in.)
Various (deflections, stackability)	± 0.2 cm (± 0.08 in.)

2.2.6 Air Transportability - Fixed Wing Internal.

Devices for Measuring	Measurement Accuracy
Scale (weight)	± 1% of reading
Tape (physical dimensions)	± 0.159 cm (± 0.06 in.)
Articulation measure (bogie articulation)	± 1 degree
Gauge (tire pressure)	± 7 kilopascal (kPa) (± 1 pounds per square inch gauge (psig))

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Devices for Measuring	Measurement Accuracy
Various (ground pressure)	± 1% of reading

2.2.7 Air Transportability - Rotary Wing Internal.

Devices for Measuring	Measurement Accuracy
Various (ground pressure)	± 1% of reading
Inclinometer (ramp angle)	± 1 degree
Scale (weight)	± 0.5% of reading
Tape (physical dimensions)	± 0.159 cm (± 0.06 in.)

2.2.8 Air Transportability - Rotary Wing External.

Devices for Measuring	Measurement Accuracy
Scale (weight)	± 0.5% of reading
Various (center of gravity location)	± 2.5 cm (± 1 in.)
Thermometer (ambient temperature)	± 2.0 °C (± 3.6 °F)
Anemometer (wind speed)	± 1 knot
Wind vane (wind direction)	± 30 degrees
Inclinometer (angle of inclination of load when suspended) (static test)	± 1 degree
Various (airspeed, angle of bank, rates of descent and climb)	As required

2.2.9 Air Transportability - Airdropped Materiel.

Devices for Measuring	Measurement Accuracy
Scale (weight)	± 1% of reading
Vertical distance (ground impact test)	± 1 cm (± 0.4 in.)
Time (rigging)	± 0.1 hour

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Devices for Measuring	Measurement Accuracy
Position (during airdrop)	± 2 m (± 7 ft)
Time (frequency response)	± 200 meters per second (m/sec) (± 656 feet per second (ft/sec))
Impact velocity	± 0.5 m/sec (± 1.6 ft/sec)
Acceleration	± 10% of reading

3. REQUIRED TEST CONDITIONS.

a. Coordinate test requirements and procedures with SDDCTEA and other applicable component transportability agencies prior to conducting any transportability tests preceding certification approval. In many cases, the representative(s) are required to witness the testing.

b. Ensure all viable transport orientations are assessed for multi-directional assets (i.e., ISO containers and shelters). If the equipment can be oriented in more than one direction for transport, the applicable restraint provisions and system design must meet the design limit loads for all possible orientations.

c. Coordinate specific test and acceptance criteria ahead of system assessment for designs incorporating bolted provision assemblies. In the case of certification testing, appropriate coordination shall occur between the material developer and applicable transport certification office to outline methods of assessment pertaining to bolted provision assemblies.

d. The use of spreader bars is not permitted for testing purpose unless approved by SDDCTEA or the appropriate component transportability agent. Ensure that stowage is provided for the spreader bars on the item as basic issue item (BII) equipment. If authorized, use spreader bars during provision testing.

e. Do not use shackles as lifting, equipment tie-down, or multipurpose provisions unless they meet the requirements of a high strength, safety (bolt/nut) anchor shackle in RR-C-271G[5] and are approved by SDDCTEA.

f. Ensure that the test item is in the designed load configuration - full payload configuration (real or simulated payload). The gross weight condition or the gross weight rating (GWR) of the test item, whichever is greater, is mandatory; other load conditions (e.g., curb weight) are as required. Use of real payload/cargo is required of most test procedures to verify survivability following the shock or vibration environment induced. Vehicles not typically transported with payload such as wreckers, truck tractors, and dump trucks are exceptions and may be tested at a curb weight or weight less than the gross weight. Consult with SDDCTEA for these types of vehicles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

g. Specific vehicle configuration standards are defined in Military and Federal standards.

(1) The following are standard definitions derived from MIL-STD-209K[6]:

(a) Gross weight is defined as the weight of the basic equipment plus the weight of any associated support items of equipment and cargo attached to, contained within, or projected as payload for the equipment (e.g., shelter). The weight of ammunition, fuel, water, and lubricant necessary to render a system combat ready are considered as payload and included in the gross weight. The gross weight equals the gross vehicle weight rating (GVWR) or the maximum projected weight of the equipment, whichever is greater.

(b) GVWR is defined as the unloaded vehicle weight, including all fluids, and its maximum payloaded (to include the weight of the crew), where the payload weight is based upon the manufacturer's rating of the structural capability of the vehicle.

(2) The Federal Motor Vehicle Safety Standard (FMVSS) 571.3[7] "Definitions" provides specific definitions for the following commercial vehicle configurations:

(a) GVWR means the value specified by the manufacturer as the loaded weight of a single vehicle.

(b) Curb weight means the weight of a motor vehicle with standard equipment; maximum capacity of engine, fuel, oil, and coolant; and, if so equipped, air conditioning and additional weight optional engine.

(c) Gross combination weight rating (GCWR) means the value specified by the manufacturer as the loaded weight of the combination vehicle.

(d) Gross axle weight rating (GAWR) means the value specified by the vehicle manufacturer as the load carrying capacity of a single axle system, as measured at the tire-ground interfaces.

(3) Other pertinent vehicle configurations may be required based on the test plan or physical circumstance. Documentation of the vehicle configuration is critical. Written and photographical documentation is required.

h. Ensure that the test item has been fully serviced (i.e., fuel, oil, coolant) or serviced to the levels specified in the rigging/tie-down procedures or applicable equipment publications. All auxiliary equipment must be installed and secured properly.

i. Ensure that Standing Operating Procedures (SOPs) for the specific test procedures are adhered to at all times. Rail impact test operations shall be conducted on track inspected for compliance with safety criteria outlined in the Unified Facilities Criteria (UFC 4-860-03).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

j. If military personnel are required, ensure a Test Schedule and Review Committee (TSARC) request is submitted within one year from the start of testing or as early as possible.

4. TEST PROCEDURES.

4.1 Lifting and Tie-down Provisions.

a. The objective of this subtest is to determine whether the lifting and tie-down provisions of military equipment comply with dimensional and/or directional limits and design, positioning, and strength requirements for transportability. Coordination with SDDCTEA prior to testing is highly recommended to ensure proper implementation of applicable military standards regarding design analysis, load calculation, test conduct, and findings analysis. Applicable military standards are outlined in Section 4, Test Procedures.

b. Method.

(1) Categorize the lifting and tie-down provisions regarding equipment type (Type I-II) in accordance with MIL-STD-209K. Inspect the provisions regarding location to determine if any interference with functioning equipment exists and that required accessibility is provided. Measure the provisions to determine compliance with dimensional requirements. Check to determine whether the working angles for equipment tie-down provisions are attainable. Specific location, clearance, dimensional and directional requirements are defined in MIL-STD-209K. System center of gravity and provision location measurements relative to a fixed position are required for calculation of applied loadings per MIL-STD-209K.

(2) Subject the tie-down provisions to calculated lateral, longitudinal, and vertical forces, as applicable, specified by MIL-STD-209K or coordinated with SDDCTEA in the case of deviation from MIL-STD-209K. Perform a static lift to establish lifting parameters (i.e., sling lengths, sling leg angles, item attitude in suspension, and apex height) for crane movements at marine terminals. Subject all lift provisions to an apex loading specified by MIL-STD-209K, or provided by SDDCTEA.

(3) Prior to load application, secure the equipment to a bedplate, deadman, or some means of attachment for anchoring purposes. Photograph and record the restraint method and equipment employed. Record measurements characterizing the applicable provisions for comparison with post-test results, as necessary. Load application will be provided by a hydraulic actuator system. Accomplish the provision loading for the durations specified in MIL-STD-209K. Measure the loads with calibrated devices such as load cells. Video observations should be used to assist in investigation of potential failure. In the case of remote operations, live video shall be used for safety observation in the case of catastrophic system failure.

(4) At the conclusion of proof loading for each applicable provision type, inspect the test item provisions for evidence of damage or failure, and verify the integrity of the supporting structure. Examine the test item for any permanent damage or deformation through visual inspection. Record and photograph any weld cracking or visible permanent deformation or set in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

the provision(s) or other structural component resulting from load application. Post-test measurements of any observations or changes in the system from its initial state should be taken to assist in investigation of potential failure or quantifying the visually observed movement or deformation.

(5) Systems designed with bolted provision assemblies will be inspected for shifting or movement of associated components. Visual indications of movement after applied loading will be referred to the material developer for further assessment. In the case of systems under test for transportability certification, the appropriate certifying office shall be contacted for guidance on acceptance criteria and a viable path forward.

(6) See MIL-STD-913A[8] for equipment requiring Helicopter Sling Load (HSL) tests (also known as External Air Transport or EAT tests). The U.S Army Redstone Test Center (RTC) provides (HSL) expertise to assess safety of proposed tests and test methods and recommends test limitations for inclusion in the Safety Release and/or detailed HSL planning. For this reason, the test center performing the HSL test must notify RTC and the U.S. Army Evaluation Center (AEC) of upcoming HSL test(s) and provide test documentation for review with ample time to support RTC concurrence and delivery of the Safety Release from AEC. The notification and HSL safety review request should include, at minimum, the information shown in Appendix E, Table E-1. A more detailed test process is provided in U.S. Army Aberdeen Test Center (ATC) SOP HSL6337.

(7) United States Navy (USN)/United States Marine Corps (USMC) equipment with internal air transport (IAT) requirements shall also comply with the USN Naval Air Systems Command (NAVAIR) Cargo Restraint Requirements.

(8) Air droppable materiel tie-downs and attachments are tested for conformance with MIL-STD-814D[9] (see Paragraph 4.9).

(9) Aircraft cargo tie-down device testing is described in MIL-DTL-25959G[10].

4.2 Rail Transportability.

The objective of this subtest is to determine whether the test item meets the specified requirements for rail transportability certification.

4.2.1 Method.

a. Rail impact testing should be conducted on track conforming to UFC 4-860-03 for railroad track maintenance and safety. It is recommended that the designated impact testing location adhere to the applicable safety inspection schedule and maintenance recommendations outlined in the standard.

b. Rail impact testing is required of all items designated for rail transport to verify the structural/functional integrity of the provisions, test item, the adequacy of the tie-down system, and the tie-down procedures. Based on the Materiel Developer's or contractor's developed rail

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

tie-down procedures, any special railcar requirements are submitted to SDDCTEA for approval and then to a test activity for rail impact testing. An engineering analysis is performed by SDDCTEA to determine if the test item's weight and physical dimensions are compatible with the Strategic Rail Corridor Network (STRACNET) and North Atlantic Treaty Organization (NATO) rail facilities. The requirements for unrestricted and restricted transport are specified in MIL-STD-1366E. The physical dimensions of the item and tie-down must fall within prescribed templates illustrated in MIL-STD-1366E. Blocking and bracing is not permitted; however, wood stanchions may be allowed if attached to trailers without prime movers. Consult with SDDCTEA if any type of wood dunnage is required. Prior to rail impact testing, MIL-STD-209K tie-down provision strength testing must be accomplished to verify the strength of the tie-down provisions; or shown through previous testing or certification that the system design is structurally sound. A summary of physical characteristic measurements to be recorded specifically for rail transport of a system are identified in Appendix A, Table A-1.

c. Prior to testing, accomplish a vehicle inspection by documenting, at a minimum, PMCS results, a checklist of items inspected (or function/operations tested), and tie-down provision disposition and condition. These inspection findings will serve as a baseline for comparison with post-test findings in results analysis.

d. Accomplish clearance measurements by measuring the outermost profile points of the test item, as secured to a railroad flatcar, using geodetic surveying equipment. The geodetic scan of the test item/ flatcar combination is then superimposed into a computer-aided design file with the base of the item set at a railcar deck height corresponding to each of the applicable rail clearance diagrams.

e. Accomplish the rail impact by securing the test item on the railcar and propelling the car into a stationary buffer car(s) using a locomotive or inclined ramp. Guidance for item restraint on a railcar is found in Modal Instruction 55-19[11] Measure actual impact speeds by determining the speed of the loaded railcar immediately prior to impact or at the point of impact, if feasible.

f. Position the stationary railcar(s) with its coupler compressed and its air and hand brakes set. Position the knuckles of the buffer and test cars for coupling.

g. Subject the moving railcar to impacts at speeds of 6.4 and 9.7 km/hr (4.0 and 6.0 mph) ± 0.8 km/hr (± 0.5 mph), and 12.9 km/hr (8 mph) + 0.8, - 0.0 km/hr (+ 0.5, - 0 mph). If the speed is less than 12.9 km/hr (8 mph), repeat the impact. Conduct the impact by accelerating the railcar to the desired speed using a locomotive equipped with a speed-measuring device or inclined ramp. Release the railcar loaded with the test item an appropriate distance from the stationary buffer railcar(s) and allow it to coast until it impacts.

h. Turn the railcar with the test item around and subject the moving railcar to an impact at a speed of 12.9 km/hr (8.0 mph) + 0.8, - 0.0 km/hr (+ 0.5, - 0.0 mph), impacting the opposite end of the test railcar from first three impacts for a total of four impacts. If it is not possible to turn the test car around the buffer car(s) because of track layout, ensure that the fourth impact is accomplished in the reverse direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

i. No adjustment of the primary rail restraints is permissible after rail impact testing commences. If an adjustment of the restraint occurs, the entire rail impact procedure must be repeated beginning at 6.4 km/hr (4.0 mph).

j. At the conclusion of each impact, inspect the test item for equipment damage, to include the tie-down provisions for evidence of damage or failure, and verify the integrity of the tie-down devices. Take photographs of the damage as needed. Once the test has started, there will be no readjustment of the test item, nor any reconditioning of the tie-down devices. If the initial tie-down devices fail, repeat the entire test provided a revised method of securing is feasible and approved by SDDCTEA. After the completion of the test, examine the test item for any permanent displacement of system components contained in or mounted on the test item, or item damage, and conduct a functional and operational checkout. Compare the results with the pretest data.

k If the test item is exposed to test conditions that exceed allowable limits, conduct an appropriate physical examination of the test item and perform an operational check (when practical) before testing is resumed.

l. If the test item can be shipped in two orientations (such as lengthwise and crosswise on the rail car), repeat the four impacts for each orientation, or have two test items on the test railcar (one mounted in each orientation).

4.2.2 Temperature Conditioning.

a. If the rail impact test is to be conducted at temperature extremes (only when facilities exist to expose the entire flatcar and test item to temperature conditioning), conduct the following prior to impacting:

(1) Secure the test item on the railcar and place the test item in the temperature chamber/enclosure and adjust the chamber air temperature to the test temperature. The temperature chamber/source should be located as close to the rail impact site as possible for best efficiency.

(2) Following temperature stabilization of the test item, maintain the storage temperature for the period specified in the test plan before rail impact testing. Temperature stabilization is attained for non-operating systems when the temperature of the functional part(s) of the test item considered to have the longest thermal lag reaches a temperature that is within the temperature tolerance of the air surrounding the test item.

(3) Disconnect the temperature conditioning chamber/source, if employed, and perform the rail impact test as stated in paragraph 4.2.1.

b. If the test item is being conditioned when detached from the railcar, conduct the following after the conditioned item is mounted on the railcar:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(1) Replace the test item in the temperature chamber and adjust chamber air temperature to the test temperature.

(2) Following temperature stabilization of the test item, maintain the storage temperature for a minimum of twice the amount of time that the test item has been out of the temperature chamber.

(3) Perform the rail impact test as stated in paragraph 4.2.1.

c. If the test item is being conditioned when attached to the railcar, but the railcar is not to be conditioned, conduct the following after the item is mounted on the railcar:

(1) Enclose the test item within the portable chamber as mounted on the railcar and adjust chamber air temperature to the test temperature.

(2) Following temperature stabilization of the test item, maintain the storage temperature for the period specified in the test plan before rail impact testing. Temperature stabilization is attained for non-operating systems when the temperature of the functional part(s) of the test item considered to have the longest thermal lag reaches a temperature that is within the temperature tolerance of the air surrounding the test item.

(3) Perform the rail impact test as stated in paragraph 4.2.1.

d. If the test item remains outside the conditioning chamber for more than 15 minutes, then reconditioning is required. If more impacts are required after 15 minutes, recondition the test item for 30 minutes after the test item has stabilized at the required test temperature.

4.2.3 Failure Definition.

a. A test item is classified as not having survived the rail impact test, and the test is deemed a failure, if any item that is attached to or included as an integral part of the test item breaks free, loosens, or shows any sign of permanent deformation.

b. The test item must be fully functional after the rail impact test. Inability of the test item to operate as designed (i.e., move, shoot, and communicate) or damage resulting in a compromise of mission essential functions will constitute a test failure.

c. Breaking of any tie-down chain will constitute a non-test. With the restraint replaced, the test must be re-started at the 6.4-km/hr (4.0-mph) impact.

d. Loosening of any tie-down chain will not constitute a test failure. Investigation to determine the reason for loose restraints is highly recommended. If any of the tie-down chains are tightened between impacts, the test must be re-started at the 6.4-km/hr (4.0-mph) impact.

e. Rail tie-down chain restraints shall be inspected for stretch or other damage after impact testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

f. Test controls should be adequate to assure repeatability of the process and maintain test parameters within prescribed tolerances. Nevertheless, any impact in which the measured speed of the test car is greater than the upper limit of the test tolerance, + 0.8 km/hr (+ 0.5 mph), is considered a valid test at that incremental impact speed provided that the materiel tested, its restraint system, and the rail car are not adversely affected (i.e., damage, permanent deformation and/or performance degradation), and the next incremental impact speed is not exceeded. This contingency does not alter the need to perform four successive impacts at progressively escalating impact speeds, thus imparting a nominally increasing amount of strain energy to the restrained test item. For example, a rail impact test in which impact speeds of 8.9, 9.7, 13.1, and 13.4 km/hr (5.5, 6.0, 8.1, and 8.3 mph) were recorded and no equipment damage was noted would be considered a satisfactory test, the 8.9 km/hr (5.5 mph) impact qualifying as the incremental 6.4 km/hr (4.0 mph) impact.

4.3 Highway Transportability.

a. The objective of this subtest is to determine whether the test item can be transported over and off highways.

b. SDDCTEA is responsible for performing an engineering evaluation of equipment designated for highway transportability. Weight and physical characteristics will be compared against requirements for highway transport. Any restrictions will be noted.

(1) Prepare the test item for shipment in accordance with the procedures contained in the appropriate technical manuals for movement over highways. Measure the physical parameters including width, height, length, gross weight, and axle wheel loading. A summary of all characteristic measurements and those specified to be recorded at a minimum for this mode of transport are identified in Appendix A, Table A-1.

(2) Any unique considerations for pintle loading should be identified, including documenting pintle loads that may exceed the permissible limit of the system design.

(3) Load, block, brace, and tie down the item on a semitrailer or other vehicle designated as the prime mover. If the technical manuals do not contain tie-down instructions, use procedures recommended by the test agency and concurred with the developing agency, where appropriate. Once the item is securely loaded, measure the physical parameters of the combination, including width, height, length, gross weight, and axle wheel loading. Record the measurements and weight.

4.4 Marine Transportability.

a. The objective of this subtest is to determine whether the test item can be transported by marine vessels. Marine transportability is composed of lifting, vessel, test item compatibility in accordance with MIL-STD-209 K, and MIL-STD-1366E, respectively. The US dry cargo fleet consists of four conventional ship types: breakbulk, container, barge carriers, and roll on/roll off (RORO). Various combinations of these four ship types also exist. The following paragraphs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

offer a short summary of each type. Specific measurements and space constraints are detailed in MIL-STD-1366E. A summary of all characteristic measurements and those specified to be recorded at a minimum for this mode of transport are identified in Appendix A, Table A-1.

b. Breakbulk General Cargo Ships. The hold configuration on most breakbulk ships is generally the same, consisting of five to seven holds. Each hold has three to five decks and hatch covers that allow access to the different decks. Cargo operations on breakbulk vessels are lift on/lift off (LOLO). Each hold on a breakbulk vessel is served by ship's gear.

c. Containerships. Modern containerships, to include combination ships, are designed to carry all or part of their cargo load in containers. The containership allows containers to be secured without the use of dunnage. Containerships also have the capability for transporting containers that are stacked on the vessel deck.

d. Barge Carriers. The Lighter Aboard Ship (LASH) and Sea Barge (SEABEE) transportation systems operate similar to a containership. In these systems, cargo is stowed in unitized lighters or barges. The barges or lighters are then stowed aboard a barge carrier or mother ship. One major difference between containerships and barge carriers is the amount of cargo that barges or lighters can handle. LASH lighters and SEABEE barges have cubic capacities of 570 and 1130 m^3 (20,000 and 40,000 ft^3), respectively. The barge carrier/mother ship is self-sustaining. The LASH system uses a gantry crane to load its lighters, while the SEABEE system has an elevator to load its barges.

e. RORO Ships. The RORO ship is primarily a vehicle transporter that allows vehicles to drive on or off the ship via ramps. RORO cargo includes wheeled, tracked, self-propelled, and towed vehicles and equipment. Most modern RORO vessels not only carry vehicles, but also carry a combination of containers and/or breakbulk cargo. A series of external and internal ramps facilitate the loading and discharge of RORO cargo. To maintain safe operations, the ramp angle for loading and unloading procedures is no greater than 15 degrees.

f. Method.

(1) Vessel and Test Item Compatibility. Upon completion of physical parameter measurements and lifting and tie-down testing in accordance with MIL-STD-209K (paragraph 4.1), vessel and test item compatibility is evaluated. Using MIL-STD-1366E, SDDCTEA determines the compatibility of the test item with marine stowage and handling provisions of the various oceangoing vessels. A comparison of test item physical characteristics with the hatch and hold dimensions and cargo handling gear is made. A summary of all general requirements and those specified to be recorded at a minimum for this mode of transport are identified in Appendix A, Table A-1. From this comparison, an estimate is made regarding the capability of the item to be transported by various types of vessels. An evaluation is also made of the capabilities of amphibious vehicles and landing craft with respect to loading and securing the test item and unloading it onto the beach. The study, depending on requirements, encompasses the physical aspects of ramp negotiation, roll on/roll off maneuvering, loading, and tie-down arrangements. Complete logistics-over-the-shore (LOTS) performance test requirements are addressed in TOP 01-2-510[12].

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(2) Cargo Movement. Conduct a trial loading of the test item on actual oceangoing vessels, using the ship's cargo handling gear or customary dockside lifts. Once loaded, stow and secure the test item using specific cargo restraints. Representative marine tie-down configurations are illustrated in Transportation Engineering Agency (TEA) Pamphlet 55-22[13]. Note the presence of test item combustibles, if any. Apply recording instrumentation, including strain gages and accelerometers, to select points of the item to record angles of pitch, roll, and yaw to correlate the environmental data logged by the ship during its voyage. When practical, inspect the test item periodically during transport in addition to the beginning and end of the voyage.

4.5 Terminals Handling and Movement.

a. The objectives of this subtest are to determine the ability of:

(1) The test item to withstand the mechanical stresses experienced when moved with MHE, in vehicles, or by personnel.

(2) The packaging and packing methods to provide protection to the contents.

b. Method.

(1) The procedures referenced in this paragraph provide simulations of the conditions typically encountered by freight containers, shelters, palletized loads, and equipment contained in transit cases or shipping crates in the course of their handling and movement at points of embarkation for and debarkation from multi-mode (rail, highway, marine, air) transportation media. A summary of all characteristic measurements and those specified to be recorded at a minimum for this mode of transport are identified in Appendix A, Table A-1.

(2) Standardization of Containers and Shelters. Intermodal freight containers and members of the Army Standard Family of Shelters (ASFS) designated as ISO shelters must be certified as meeting the requirements for multi-mode transport, or be tested to ISO Standard 1496/1:2013 (paragraph 1 of this TOP for ANSI/ISO test facilities information) and satisfy the engineering and design requirements of American Society of Testing and Materials (ASTM) E1925-10[14].

(3) Forklift Handling. The adequacy of the forklift provisions of shelters, pallets, and shipping crates, and their ability to be safely lifted and restrained during movement by forklift without experiencing more than superficial damage is determined in this procedure. Given the variety of MHE types available at transportation terminals, any forklift of sufficient capacity and tine characteristics (load center) may be used; tactical, conventional, or non-DOD. The impact of forklift handling on the test item, not the ability of accepted MHE to handle its load, is the issue, thus allowing this flexibility in MHE application. Inspect the item to identify the forklift pockets, and determine any recommendations and restrictions listed on the data plates. The size of most shelters necessitates the use of forklifts with tines of at least 2.4 m (8 ft) in length or the application of tine extenders. The procedure for testing shelters is contained in ASTM E1925-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

10, paragraph 10.5, Forklift Handling Test; while ASTM D6055-07[15] and ASTM D6179-07[16] provides a means of testing unitized loads and large shipping cases and crates. In each case, load the test item to its gross shipping weight (GSW), with center of gravity as near as practicable to the actual service conditions, and affix the load to the forklift with slack safety chains or cabling to minimize the possibility of loss of the load.

(4) Towing and Dragging. The ability of shelter skids, pallets, and shipping crates to withstand lateral loads incurred during placement adjustments of these items and their movement in the absence of forklifts is determined. Load the test item to its GSW, with center of gravity as near as practicable to the actual service conditions. Shelters with skids should be tested in accordance with ASTM E-10, paragraph 10.28; and pallets and shipping crates with skids are presented in ASTM D6055-07, Test Method D, paragraph 9.4. Connect a load cell to the towing medium in order to measure and thus control the applied force.

(5) Lifting. The adequacy of lifting provisions and the structural integrity of the test item when lifted are examined in this test. Test procedures for lifting in general and for external (rotary wing and tiltrotor aircraft) air transport are discussed in paragraphs 4.1 and 4.8, respectively, of this TOP. Procedures for the testing of shipping crates are provided in ASTM D6055-07, Test Method E, Grabhook Test Procedures, paragraph 10-1; and Test Method F, Sling Test Procedure, paragraph 10.2. Intermodal freight containers shall be certified as compliant with ISO Standard 1496/1:2013 or tested to this document.

(6) Drop Tests.

(a) The structural integrity of the test item and the effectiveness of any internal shock attenuation mechanisms are determined in this series of tests. Procedures relevant to handling of shipping crates and palletized loads (except for munitions) such as corner drop, edgewise drop, and free fall drop are found in the following documents:

1 Rotational Corner Drop Test ASTM D6179-07, Test Method B.

2 Rotational Edge Drop Test, ASTM D6179-07, Test Method A.

3 Unsupported Free Fall Drop Test, ASTM D6179-07, Test Method D.

4 Rotational Flat Drop Test, ASTM D6179-07, Test Method C.

(b) The free-fall test is limited to packages or containers of up to 70 kg (150 lb) in weight and not exceeding 152 cm (60 in.) on any edge or diameter, whereas the other tests of ASTM D6179-07 are intended for materiel with physical characteristics exceeding these parameters. Separate procedures for flat (free-fall) and rotational (edgewise) drop testing of shelters both with and without skids are contained in ASTM E1925-10, paragraphs 10.29 (without skids) and 10.30 (with skids). Procedures for drop testing of munitions are available in International Test Operations Procedure (ITOP) 04-2-601[17]. Each of these tests employs a quick-release mechanism at the apex of the rigging to release the test item from its point of suspension at the initiation of the drop. If the test item has no provisions for protection from impact by

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

falling rigging, the sling set is tethered to the overhead crane hook such that it neither strikes the test item nor interferes with its motion in dropping. The test item may be instrumented to measure mechanical shock and vibration, and strain incurred by selected components or members. Any shock and vibration measurements of internal components or members should be accompanied by comparable recordings at the base of the test item (forcing function).

(7) Rough Handling. This procedure is used to determine the capability of the test item to withstand the shocks normally induced by loading and unloading of equipment in routine service usage, not to simulate handling the logistics shipping environment. Measurement of shock and vibration imparted to the test item must be consistent with the guidance provided for drop testing (paragraph 4.5b(6)); however, the application of instrumentation may not be feasible in the case of small items. The guidance provided in MIL-STD-810H[18], Test Method 516.8, Procedure IV, Transit Drop; is applicable for equipment packaged in its transit case(s) as prepared for field use. Simulation of conditions experienced by items carried by individuals on their person or as unsecured cargo in trucks is described in ITOP 04-2-602[19]. Munitions, rifles, rockets, radios, and mortars are covered in this latter document.

(8) Cargo Compatibility. Methods for determining the susceptibility of vehicles for loading and unloading of cargo are found in TOP 02-2-537[20]. This document prescribes procedures for trucks, aircraft, ships, and railroad car carriers. These tests can usually be integrated with other terminals handling and movement procedures.

(9) Stackability. The ability of shipping crates or packages to withstand loads such as these imposed on the bottom container in a stack, or support loads on their tops, with or without dunnage present, is determined through the use of the test procedures of Superimposed Load Test; Stackability, with Dunnage , and Superimposed Load-Test; Uniformly Distributed, without Dunnage as prescribed in ASTM D4169-09[21]. Strain measurements of stressed members may be taken if required.

4.6 Air Transportability - Fixed Wing Internal.

a. The objective of this subtest is to determine the suitability of the test item to be transported by United States Air Force (USAF) and Civil Reserve Air Fleet (CRAF) fixed-wing aircraft.

b. Method. Certification of military equipment for transport on USAF and CRAF aircraft is performed by the USAF Air Transportability Test Loading Activity (ATTLA) through coordination with SDDCTEA. The required data for each test item are recorded and forwarded to ATTLA through SDDCTEA.

(1) Item Inspection and Physical Characteristics. Inspect the item and record the location of, and data on, all tie-down points. The tie-down fixtures must conform with the design criteria of MIL-STD-209K. Measure and record the weight and physical characteristic measurements of the item as illustrated in Figures 1 through 3. A summary of all characteristic measurements and those specified to be recorded at a minimum for this mode of transport are identified in Appendix A, Table A-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

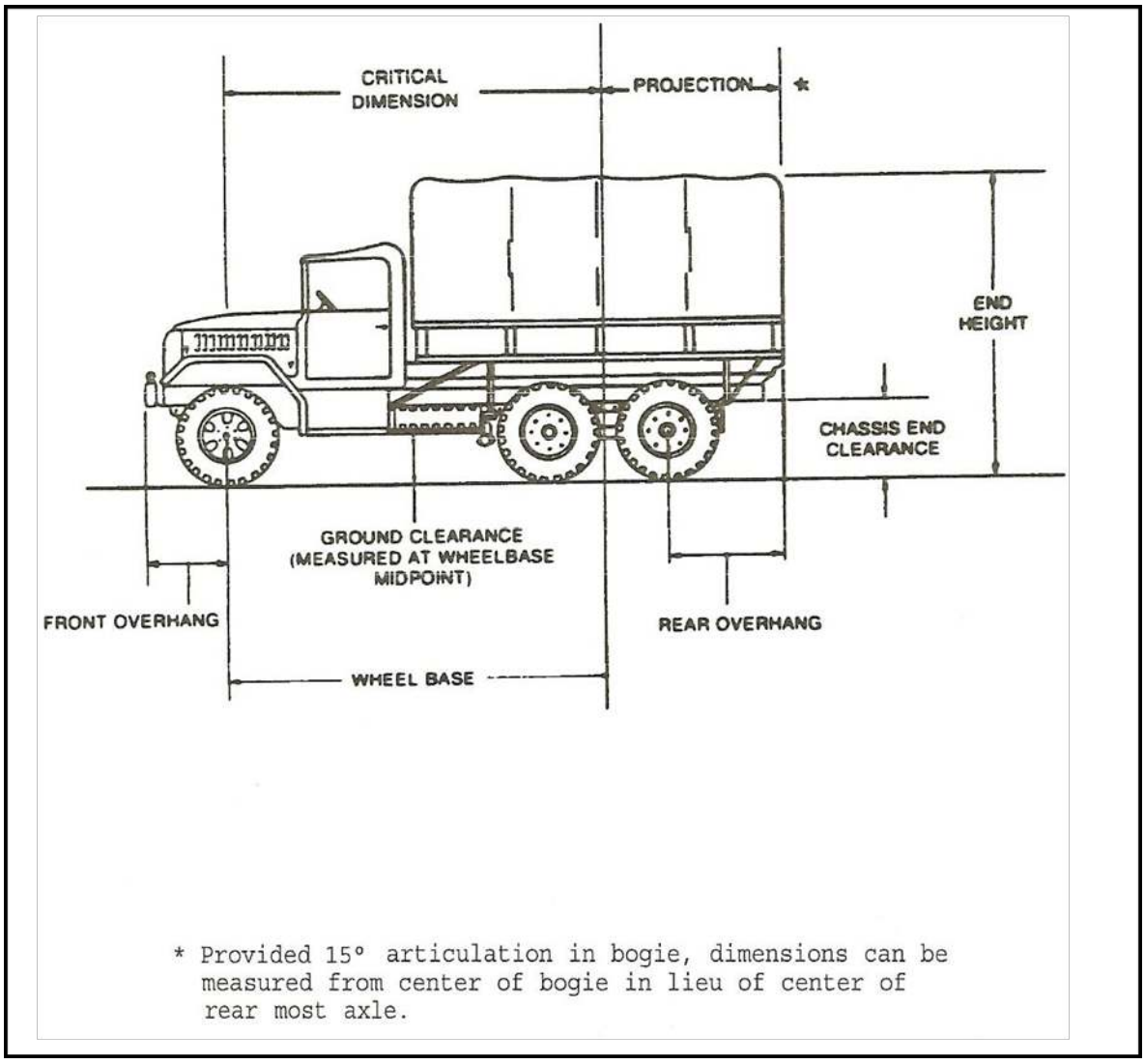


Figure 1. Illustration for determining dimensions of physical characteristics for a truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

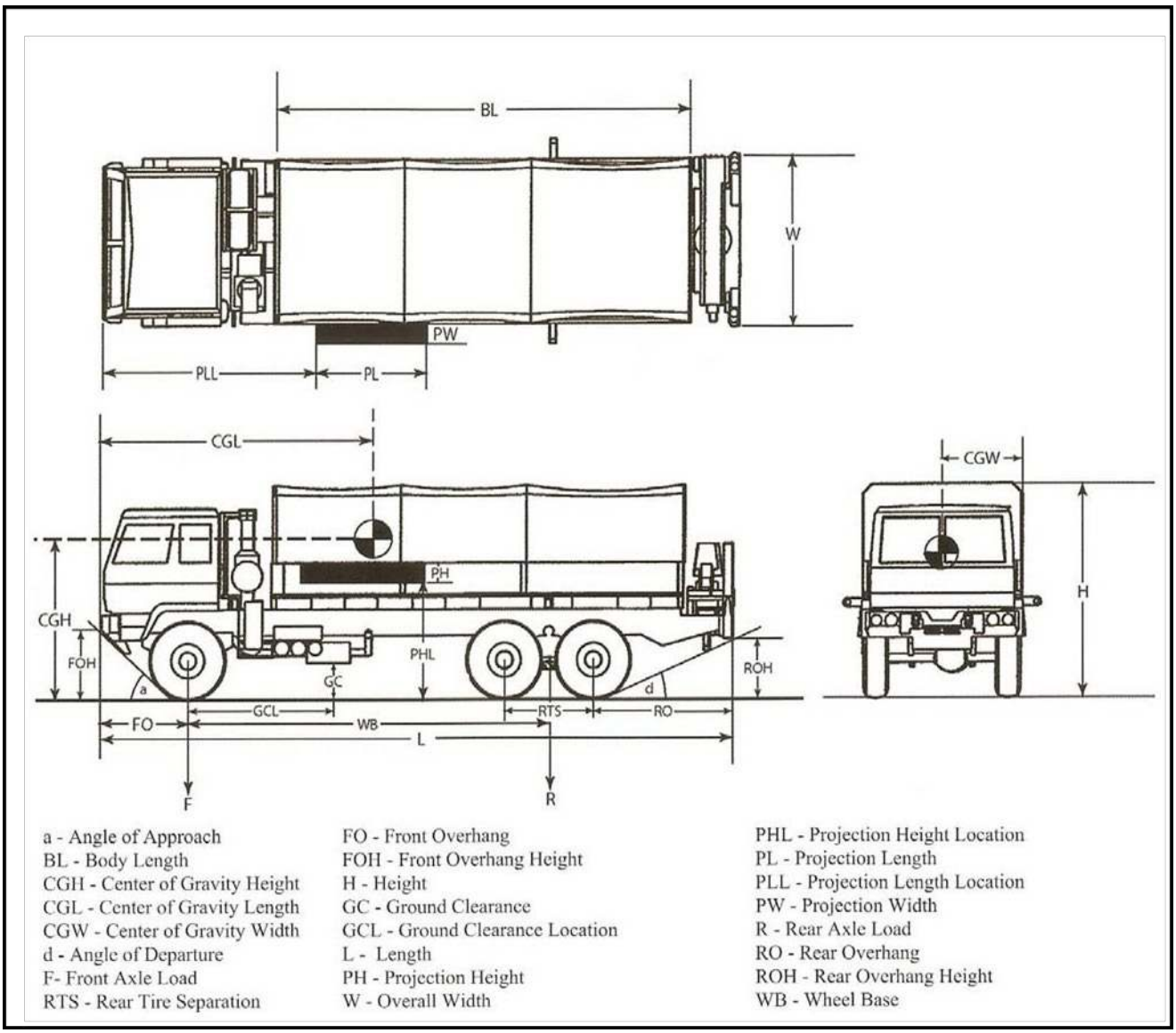


Figure 2. Illustration for determining dimensions of physical characteristics for a truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

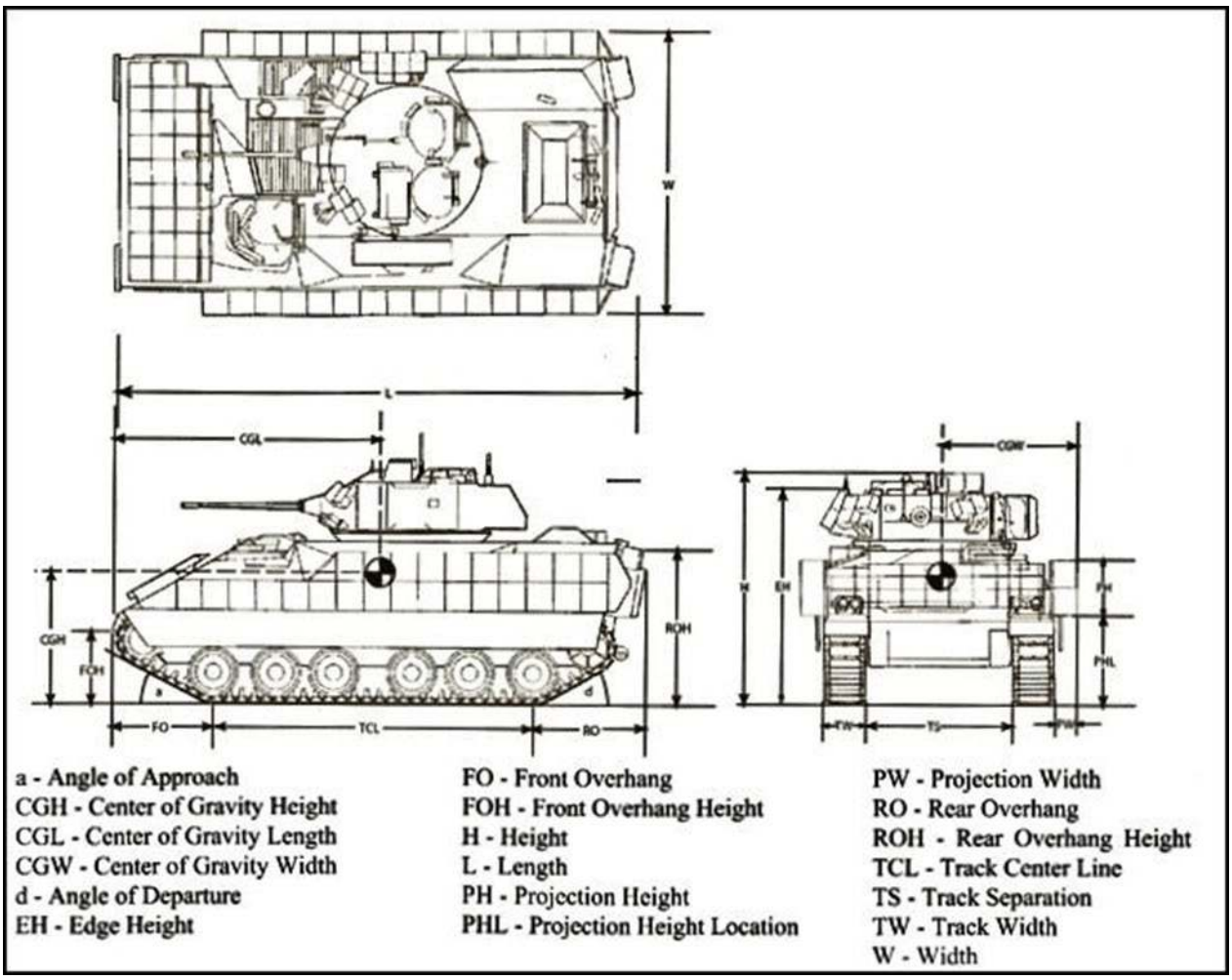


Figure 3. Illustration for determining dimensions of physical characteristics for a track vehicle.

(2) Forward the data recorded in paragraph 4.6b through SDDCTEA to ATTLA. ATTLA will perform an analysis of the test item measurements in accordance with MIL-STD-1791C[22] to determine the item's suitability for transport on USAF aircraft. Approval status is based upon an engineering evaluation of load characteristics and tie-down provisions, proof load testing of applicable provisions, and a validation test loading, when deemed necessary. ATTLA provides the engineering analysis results and approval to USAF Air Mobility Command (AMC), USAF Air Combat Command (ACC) or USAF Special Operations Command (AFSOC), which are responsible for certification of material. If ATTLA determines that there are no significantly limiting factors involved in loading the item on all aircraft in the USAF fleet, they will issue a letter through SDDCTEA stating such. Should ATTLA determine that the test item or aircraft may be damaged during loading or transport of the item in aircraft, they will require a test loading of the item

(3) When required, coordinate test loading with ATTLA and the USAF operating commands. These agencies will oversee the test with particular attention paid to those factors

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

highlighted as potential hazards to the aircraft and the test item during the analysis. All test loadings are conducted in accordance with MIL-STD-1791C.

4.7 Air Transportability - Rotary Wing/Tiltrotor Internal.

a. The objective of this subtest is to determine the capability of the test item to be transported in an internal configuration by rotary-wing aircraft. The Aerial Delivery Engineering Support Team, FCDD-SCD-SAE, of the U.S. Army DEVCOM SC is responsible for certification of equipment transported internally via U.S. Army rotary-wing aircraft. NAVAIR has been designated for USN and USMC internal load certifications. The DEVCOM SC certification is based on an engineering evaluation of load characteristics and the tie-down provisions accomplished by DEVCOM SC , proof load testing of the tie-down provisions in accordance with MIL-STD-209K and a validation test loading, when deemed necessary. The DEVCOM SC engineering evaluation includes the following.

b. Identification of the helicopters that are capable of transporting the test item based on each helicopter's size and weight limitations specified in MIL-STD-1366E and provided as Appendix B.

c. Determination of restraint criteria based on aircraft requirements specified in MIL-STD-1366E and provided as Appendix B.

d. Calculation of tie-down provision proof loads.

e. Evaluation of load weight distribution versus aircraft floor and/or ramp loading limitations specified in individual helicopter technical manuals.

f. Determination of item clearances during loading and flight. An item must maintain 15 cm (6 in.) of vertical clearance and 13 cm (5 in.) of clearance on each side of the aircraft structure at all times during loading and flight.

g. Method.

(1) Test the tie-down provisions in accordance with MIL-STD-209K. Compliance with MIL-STD-209K satisfies the crash load restraint criteria for military rotary- wing aircraft with the exception of the U.S. Army UH-60. The UH-60 12-*g* forward crash load restraint criterion will require additional testing or equipping the item(s) with a greater number of forward tie-downs. Each tie-down provision and its supporting structure shall withstand its proportionate share of the loadings. All tie-down provisions must undergo proof load testing. No visible permanent deformation or set in the provisions or supporting structure shall result from the application of the loads. In addition, the contractor or manufacturer will provide a material analysis illustrating that the ultimate load is not less than 1.5 times the required design limit load for the provisions.

(2) Conduct test loading of the item if the DEVCOM SC engineering analysis reveals that the characteristics of the test item are such that analytical means alone are insufficient to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

determine whether it meets the clearance, projection, overhang, and ramp cresting requirements. Load the test item in the designated aircraft and secure it in accordance with the procedures outlined in Field Manual (FM) 55-450-2[23]. During the test loading, document and photograph any special procedures for loading and/or tie-down developed. If any restrictions are revealed, attempt to disassemble the item to enable loading and document the disassembly required.

4.8 Air Transportability - Rotary Wing/Tiltrotor External.

a. The objective of this subtest is to determine the capability of the test item to be transported as an externally suspended load by rotary-wing and/or tiltrotor aircraft.

b. Method.

(1) Inspect the test item lift provisions to determine the locations, type, and measurements of the provisions for conformance to MIL-STD-913A. Weigh the item and determine the center of gravity. Communicate this information to the DEVCOM SC (FCDD-SCD-SAE) which is responsible for performing an engineering analysis and developing draft rigging procedures. The aircraft that are capable of transporting the item are identified in the engineering analysis. The maximum external payload capabilities of U.S. Army, USN, and USMC rotary-wing and tiltrotor aircraft are specified in MIL-STD-1366E and provided in Appendix C. A prerequisite to a flight demonstration is the successful accomplishment of proof load testing of the lift provisions and/or compression testing of the item in accordance with MIL-STD-913A, described previously in paragraph 4.1. Prior to a flight demonstration, inspect all lift provisions and adjacent structural components to ensure integrity. If any damage or deformation exists, the item should not be flown pending repair or replacement.

(2) The test item shall undergo static lift testing to verify sling leg clearances and load orientation prior to a flight demonstration. Rig the item with a standard military sling set in accordance with the draft rigging procedures provided by DEVCOM SC and in accordance with Technical Manual (TM) 4-48.09[24]. The sling set nomenclature and capacities are listed in Table 1.

TABLE 1. MILITARY SLING SETS FOR EXTERNAL AIR TRANSPORT

SERVICE	SLING SET CAPACITY		NSN	SLING LEG LOAD LIMIT		TYPE
	kg	lb		kg	lb	
Army	4,536	10,000	1670-01-027-2902	5,126	11,300	Rope
Army	11,340	25,000	1670-01-027-2900	10,206	22,500	Rope
USMC	18,144	40,000	3940-01-183-2118	18,053	39,800	Rope
USN and USMC Vertical/Short Take-Off and Landing (V/STOL V-22/ MV-22 Aircraft)	11,340	25,000	1670-01-027-2900	10,206	22,500	Rope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(a) Employment of a spreader bar to avoid sling-to-item interference is discouraged due to the logistics burden and additional rigging time required. A spreader bar used for external air transport must be stowed with the test item as On-Vehicle Equipment (OVE) or a BII. Additionally, compressive load testing of the spreader bar, unless previously tested or type classified will be required prior to flight testing.

(b) Once rigged, attach the sling set apex to the cargo hook of a mobile or overhead crane and hoist the item from the ground. Note any sling-to-item contact or interference. Measure and record the angle of inclination of the load with respect to the ground. If the desired orientation identified by DEVCOM SC was not achieved (i.e., level, 5 degree nose downward), adjust the rigging until the desired orientation is attained. Record the final rigging configuration (link counts).

(3) Position the test item on the test pad with its designated front facing into the prevailing wind. Engage the parking brakes on vehicles, trailers or weapon systems equipped with parking brake systems. Secure loose brake lines, inter-vehicular cables and safety chains of towed items with tape or nylon cord. Remove, stow, and secure all soft-top covers in a designated shelter or to the item in some fashion. Secure all external and internal equipment, and vents and access doors using tape, nylon cord, or cargo straps if a positive means of securement is not already provided. Rig the sling set in a breakaway fashion to prevent any interference during load hookup. Photograph the resultant rigging configuration.

(4) Fly the item as an externally suspended load with each military aircraft specified and for each rigging configuration desired (i.e., single and dual point rigging). The rigging crew will access the highest point of the load. Document any difficulties experienced in accessing or departing the load due to a lack of adequate footholds, handles, etc. As the aircraft hovers over the load, the static discharge wand handler contacts the wand (if required) to the cargo hook of the aircraft as the second rigger attaches the apex fitting to the cargo hook. Once the rigging crew has departed the load, the aircrew performs the flight demonstration in accordance with the process listed in the Multi-Service Flight Data Collection Sheet (MSFDCS), provided as Appendix D. The aircraft will accomplish the flight demonstration in a predetermined flight path/area that is predominately uninhabited. Photograph the test item slung under each aircraft to document load orientation. Record (video) the flight demonstration by chase aircraft. After the completion of the flight demonstration, the aircrew returns the load to the test pad. Perform visual and functional inspections of the item. Record and photograph any evidence of physical damage as a result of the flight demonstration.

4.9 Air Transportability - Airdropped Materiel.

a. The objective of this test is to determine the suitability of the test item to be airdropped from fixed-wing aircraft. DEVCOM SC is responsible for certification of Army equipment for airdrop. DEVCOM SC is recognized in the DOD as the subject matter expert for cargo airdrop. And as a result, the other services request that DEVCOM SC support their cargo airdrop certification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

b. Method.

(1) Lift/Tie-down Provision Compatibility. Inspect the test item for location and compatibility of tie-down and lifting points. Employ data obtained from the lifting and tie-down provisions subtest (paragraph 4.1) when appropriate. Verify compatibility of test item points with load anchoring points on the aircraft or airdrop platform. Inspect and measure tie-down, suspension, and extraction provisions to identify their number, location, dimensions, and clearances as required by MIL-STD-814D.

(2) Provision Integrity and Aircraft Compatibility. Prepare the test item for airdrop by either the suspension or extraction method, or both, as identified by DEVCOM SC in accordance with TM 10-500-2[25] and MIL-HDBK-669[26]. Weigh and measure overall physical characteristics of the rigged load to determine conformance with MIL-STD-814D and for compatibility with the designated carrier aircraft. Accomplish static pull tests of suspension or extraction provisions and components to verify compliance with MIL-STD-814D (paragraph 4.1).

(3) Energy Dissipation. If specific provisions for energy dissipation are not provided, subject the prepared test item to deceleration force levels less than the *g* +1, or 19.5 times the item airdrop weight, specified by MIL-HDBK-669, using trial force levels as recommended by the developer. Progressively increase low force levels, making observations for any indications of damage, up to the maximum ratio of *g* +1 or 19.5 ± 10 percent. However, the energy dissipation kit should be provided by DEVCOM SC.

(4) Simulated Airdrop Impact Test (SAIT) (Low Velocity). Assemble, secure, balance, and cushion the test item on a pallet or other appropriate carrier in accordance with DEVCOM SC draft procedures and MIL-HDBK-669. Attach the test item to the cargo lifting hook of a static drop facility. The facility and lifting device will be high enough to permit the bottom of the cargo to be raised a minimum of 3.87 m (12.7 ft), the equivalent of 8.7 m/sec (28.5 ft/sec) free-fall, above the ground. The impact area will be level, constructed of a minimum of 0.3-m (1.0-ft) thick reinforced concrete or similar rigid material. Affix an accelerometer(s) to a centrally located hard point and/or to any sensitive items within the load. Attach the rigged item to the lifting hook of the facility and raise the item off the ground until its lower edge is positioned 3.87 m (12.7 ft) above the ground. Adjust the rigging material to achieve a level suspension attitude, as necessary. Record the vertical distance from the ground to the load. Release the cargo hook, allowing the item to free-fall to the ground. Record the accelerometer readings. Conduct visual and functional inspections of the load and record the results.

(5) Airdrop. Prior to live airdrop testing, an airdrop Proposed Test Plan (PTP) must be submitted to the U.S. Air Force Aeronautical Systems Center (HQ ASC) at Wright-Patterson AFB, and U.S. Air Force Air Mobility Command (HQ AMC) at Scott AFB. A PTP approval from HQ ASC and HQ AMC is required before live airdrop testing can be conducted from Air Force fixed wing aircraft. Rig the test item in accordance with instructions provided/approved by DEVCOM SC. Document any deviations or additions to the instructions. After completion of the rigging, conduct a Joint Aircraft Inspection (JAI) to verify procedures and load worthiness. Transfer the load to the aircraft by any safe means available and acceptable to the aircrew. Document the load transfer procedure, including the identification of aircraft loader employed,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

when applicable. Secure the item to the aircraft in accordance with USAF instructions. Document any deviations or additions to the instructions. Following item securement, conduct a second and final JAI. Upon completion of the JAI and acceptance of the load, all responsibility for the mission is transferred to the USAF. The altitude, air speed, and drop zone for the test are identified and coordinated with the air crew, test facility, and USAF. Employ ground tracking, photography, and instrumentation to record position versus time, flight trajectory, and impact velocity and acceleration. After load impact, conduct a visual inspection of the test item, instrumentation and parachute system (airdrop only). Record and photograph any evidence of physical damage as a result of the airdrop. After the completion of de-rigging, conduct a visual and functional inspection of the load. Record the results of the inspections.

5. DATA REQUIRED.

5.1 Lifting and Tie-down Provisions.

a. Dimension and location measurements of lifting and tie-down provisions, number of provisions, and directional capabilities.

b. Record of any sling-to-item interference or provision accessibility problems.

c. Description of procedure employed in securing test item for load application.

d. Measurement of static force(s) applied, the direction of each static pull, and any physical changes in the lifting/tie-down provisions(s) subjected to the pull.

e. Measurement of the sling lengths, sling leg angles, inclination, and apex height of the suspended item.

f. Identification of spreader bar(s) employed, as necessary, and the configuration of the lifting assembly (photographs when available).

g. Record of any visual permanent deformation of the provision or supporting structure, to include settling or weld cracking resulting from testing (photographs when available).

h. Record of any quantifying measurements supplementing the observed visual deformation.

i. Record the weight and center of gravity applicable to transport configuration.

5.2 Rail Transportability.

a. Documentation of the tie-down devices used to secure the test item (photos, drawings, materials list, etc.).

b. Description of any existing damage of test item or railcars.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

c. Description of the test railcar and individual buffer car(s).

d. Description of the test item, including configuration, loading, weight, dimensions, tire pressures, and general location on the test car.

e. Documentation of results for pre-test system inspection (to include any functional or operational system checks).

f. Direction of the impact (define forward end of test item).

g. Impact speeds.

h. Results of post-test examination and operational checkout tests.

i. If necessary, transducer measurements (i.e., acceleration, temperature, strain, etc.) of various locations on the test item and railcar.

j. Effects on the test item and tie-down devices including deficiencies, shortcomings, or limitations that occur during rail impact.

k. Locations and extent of any interference with railroad clearance diagrams.

5.3 Highway Transportability.

a. Weights, gross, curb, and axle loadings of test item and carrier supporting the test item.

b. Configuration and any loading of the test item for highway transportation.

c. Physical dimensions (including length, width, reducible width, overall height, and reducible height) of the test item.

d. Physical dimensions (including overall length, width, height, and any overhang(s)) of the test item as secured on its carrier.

e. Description of the tie-down devices used to secure the test item to its carrier (photos, drawings, materials list, etc.).

5.4 Marine Transportability.

a. Type of ship or simulation gear used.

b. Angles of approach and departure for test vehicle.

c. Ability of test vehicle to negotiate a 15-degree ramp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

d. Minimum ground clearance of test vehicle (identify location).

e. Equipment used in loading and any difficulties encountered.

f. Location of stowage.

g. Method of securing item.

h. Length of time and number of people required to rig the test item for shipment.

i. Measurements of acceleration, deceleration, and strain.

j. Duration of simulation or voyage, and angles and periods of pitch, roll, and yaw encountered.

k. Condition of bracing and securing gear during and after the voyage.

l. Direction of shift, if any.

m. Amount and type of damage, if any.

n. For landing craft trials, vehicle and craft ramp angles of approach, departure, break and crest.

o. For landing craft trials, gradeability capabilities, or limitations of the test item.

5.5 Terminals Handling and Movement.

a. Results of visual inspections, dimensional records, and operational checks, the latter if appropriate, for pre- and post-test examinations.

b. Descriptions and photographs or diagrams of loading the test item, to include any blocking, bracing, or cushioning.

c. Descriptions and photographs of the test facility, including means of item handling and/or lifting, and sling capacity.

d. Measurement of heights of drop tests, and the orientation of the item for each test iteration.

e. Dimensions, type, location, configuration, and number of skids and/or forklifting provisions.

f. Description and photograph of any incompatibility of the container, shelter, or package with its storage or transportation medium.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

g. Shock and vibration, or strain data.

h. Video record of forklifting, drop, and rough handling tests.

i. Gross and tare weights of the test item.

j. Prevailing meteorological conditions at the time of the test.

k. Measurements of any deflections obtained during stackability tests.

5.6 Air Transportability – Fixed Wing, Internal.

a. Center of gravity of test item.

b. Maximum ratings of lift and tie-down provisions as determined by physical testing conducted in accordance with MIL-STD-209K.

c. All measurements contained in Appendix A, Table A-1.

5.7 Air Transportability - Rotary Wing/Tiltrotor, Internal.

a. Item description and nomenclature.

b. Weight, dimensions, center of gravity, and cubage of item.

c. Ground contact area (i.e., skids, tires).

d. Calculation of ground pressure = weight and ground contact area.

e. Number, location and load capacity and rating of the tie-downs.

f. Instructions for special servicing and preparation.

g. Orientation for flight, when critical.

h. Identification of cargo handling equipment employed, if any.

i. Precautions and procedures to be observed during loading or unloading.

j. Restrictions and clearances for loading, including ramp.

k. Reduction in vehicle and trailer tire pressure to meet floor loading requirements, if necessary.

l. Any disassembly of the item required for loading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

m. List of tie-down equipment employed.

5.8 Air Transportability – Rotary Wing/Tiltrotor, External.

a. Record of test item condition for test (fuel and lubricant levels, missing, or damaged components).

b. List of test item preparation for flight testing.

c. Configuration and tie-down diagrams for pallet loads, if applicable.

d. Sling set type and number of sling legs employed.

e. Sling set link counts, front and rear.

f. Characteristic photographs of rigged test item.

g. Angle of inclination of the load with respect to the ground during static testing.

h. Apex height (of at least one configuration) during static lift assessments.

i. Record of any difficulties experienced in accessing and departing test item by rigging crew.

j. Photographs of test item slung from each aircraft.

k. Videotape of flight test, when available.

l. Completed and signed MSFDCS, including item weight and prevailing meteorological conditions.

m. Results of post-test visual and functional inspections.

5.9 Air Transportability – Airdropped Materiel.

a. Dimensional and strength measurements of tie-down, lifting and anchoring points, number, and location.

b. Rigging diagrams and procedures.

c. Weight and overall physical characteristics of rigged item.

d. Compatibility of the test item with the airdrop platform and the aircraft.

e. Static proof load test results (paragraph 4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

f. Energy dissipation test results.

g. Vertical distance between item and ground (ground impact test).

h. Free-fall acceleration (ground impact test).

i. Time to rig test item.

j. Photographs of the rigged configuration.

k. Position versus time data at a rate of five times per second (airdrop).

l. Impact velocity (airdrop).

m. Impact accelerations (airdrop).

n. Results of visual and functional inspections.

o. Video and high-speed video of SAIT.

p. Video for live airdrop.

6. PRESENTATION OF DATA.

6.1 Lifting and Tie-down Provisions.

Compile a list or diagram specifying the location and classification of each lifting and tie-down provision, if a data plate or similar information has not been provided by the materiel developer or manufacturer. Describe the method and equipment employed to secure the item for each load application. Tabulate the results of the pull testing. Any deformation or weld cracking of the provisions or adjacent structural components is considered a failure to satisfy the Military Standard. Photographs will document the physical damage or deformation experienced. Close up photographs will document size and location of lift and tie-down provisions. Sample formats for collecting data are provided in Tables 2 through 7.

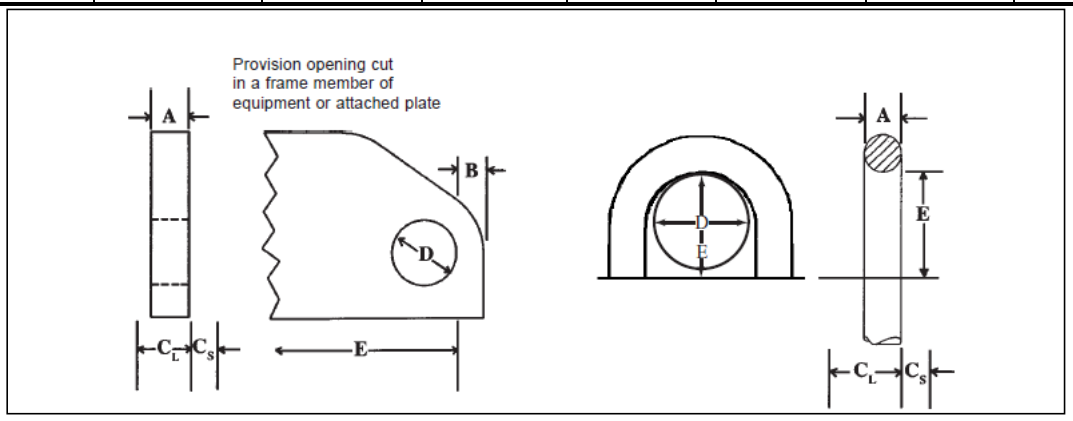
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 2. SAMPLE TABLE FOR DISPOSITION OF PROVISIONS

Provision Location	Fitting Type	Function	No. of Provisions							Labeling (Stenciled Text)	Stenciled Text Greater Than 1 in. Height
			Total	Direction of Restraint							
				Forward	Rearward	Left	Right	Vertical	Lift		
Front	Padeye	Lift									
Rear	Padeye	Lift									
Front	Padeye	Tie-down									
Rear	Padeye	Tie-down									
Total Provisions											

TABLE 3. SAMPLE TABLE FOR LIFTING AND EQUIPMENT TIE-DOWN PROVISION DIMENSIONS

Provision	Required, in., For Equipment Weight From 0 to <10,000 lb						
	Maximum		Minimum				Height
	A	B	C_L	C_S	D	E	
	1	1	7	3	3	3.2	24<h<72
Equipment tie-down							
Left Front							
Right Front							
Right Rear							
Left Rear							
Multi-purpose							
Left Front							
Right Front							
Right Rear							
Left Rear							
Lift							
Left Front							
Right Front							



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 4. SAMPLE TABLE OF CHARACTERISTICS FOR LIFT AND TIE-DOWN PROVISIONS

Parameter	
Gross weight, lb	
Lateral distance from center of gravity (CG) (equipment tie-down)	
Right front tie-down provision, in	
Left front tie-down provision, in	
Right rear tie-down provision, in	
Left rear tie-down provision, in	
Longitudinal distance from CG (equipment tie-down)	
Right front tie-down provision, in	
Left front tie-down provision, in	
Right rear tie-down provision, in	
Left rear tie-down provision, in	
Longitudinal distance from CG (lift)	
Front lift provisions, in	
Rear lift provisions, in	
Height from ground (lift)	
Front lift provisions, in	
Rear lift provisions, in	
Lateral distance from CG (lift)	
Left front lift provision, in	
Right front lift provision, in	
Right rear lift provision, in	
Left rear lift provision, in	

TABLE 5. SAMPLE TABLE OF LIFT PROVISION PROOF LOAD CHARACTERISTICS

Parameter	Value	
HSLWT, lb		
MPFA, ft^2		
MLPLF		
Required apex loading, lb		
Parameter	Rigging	
	Front	Rear
Link count of replicated sling leg		
Number of chain extensions		
Parameter	Value, deg	
Achieved inclination of restrained test item		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 6. SAMPLE TABLE OF LIFT PROVISION PROOF LOAD SUMMARY

Provision	Load, lb			
	Design	Applied Load		
		Average	Maximum	Minimum
Apex				
Left front	NA[a]			
Right front				
Right rear				
Left rear				

TABLE 7. SAMPLE TABLE OF TIE-DOWN PROVISION PROOF LOAD SUMMARY

Axis Of Loading	Type Of Loading	Provision	Load, lb			
			Design	Applied Load		
				Average	Maximum	Minimum
Lateral	Applied					
	Reaction					
	Applied					
	Reaction					
Longitudinal	Applied					
	Reaction					
Vertical, Down	Applied					
	Applied					

6.2 Rail Transportability.

a. Tabulate the test configuration and speeds of the rail impact. Prepare photographs and/or sketches of tie-down configurations. Overlay the test item profile on each applicable rail clearance diagram with clearance dimensions and railcar height. Prepare photographs of setup, instrumentation, and any damaged components. Sample formats for collecting data are provided in Tables 8 through 10, and Figure 4.

TABLE 8. SAMPLE TABLE OF RAIL IMPACT TEST CONFIGURATION

Parameter		Value
Restraint, Quantity		
Restraint, Type		
Restraint, Size, inch		
Test Weight, lb		
Tire pressure, pounds per square inch (psi)	Axle 1	
	Axle 2	
Gear Selector Position		Neutral
Parking Brake		Disengaged

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 9. SAMPLE TABLE OF RAIL IMPACT TEST SPEEDS

Impact No.	Speed, mph		Orientation
	Objective	Actual	
1	4 ± 0.5		Forward
2	6 ± 0.5		Forward
3	8 + 0.5		Forward
4	8 + 0.5		Reverse

TABLE 10. SAMPLE TABLE OF A RAILROAD RAILCAR CLEARANCE SUMMARY

Railroad System	Clearance Diagram	Railcar Type	Deck Height, in.	Deck Height, mm	Within Clearance
NATO	GIC	RS Railion	51.4	1305	
NATO	Envelope M	RS Railion	51.4	1305	
CONUS	AAR	DODX 40000	49.0	1245	
CONUS	AAR	DODX 41000	50.0	1270	
CONUS	AAR	DODX 42000	51.0	1295	
CONUS	STRACNET (DOD)	DODX 40000	49.0	1245	
CONUS	STRACNET (DOD)	DODX 41000	50.0	1270	
CONUS	STRACNET (DOD)	DODX 42000	51.0	1295	
Korean	Korean	Korean 70T	55.1	1400	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

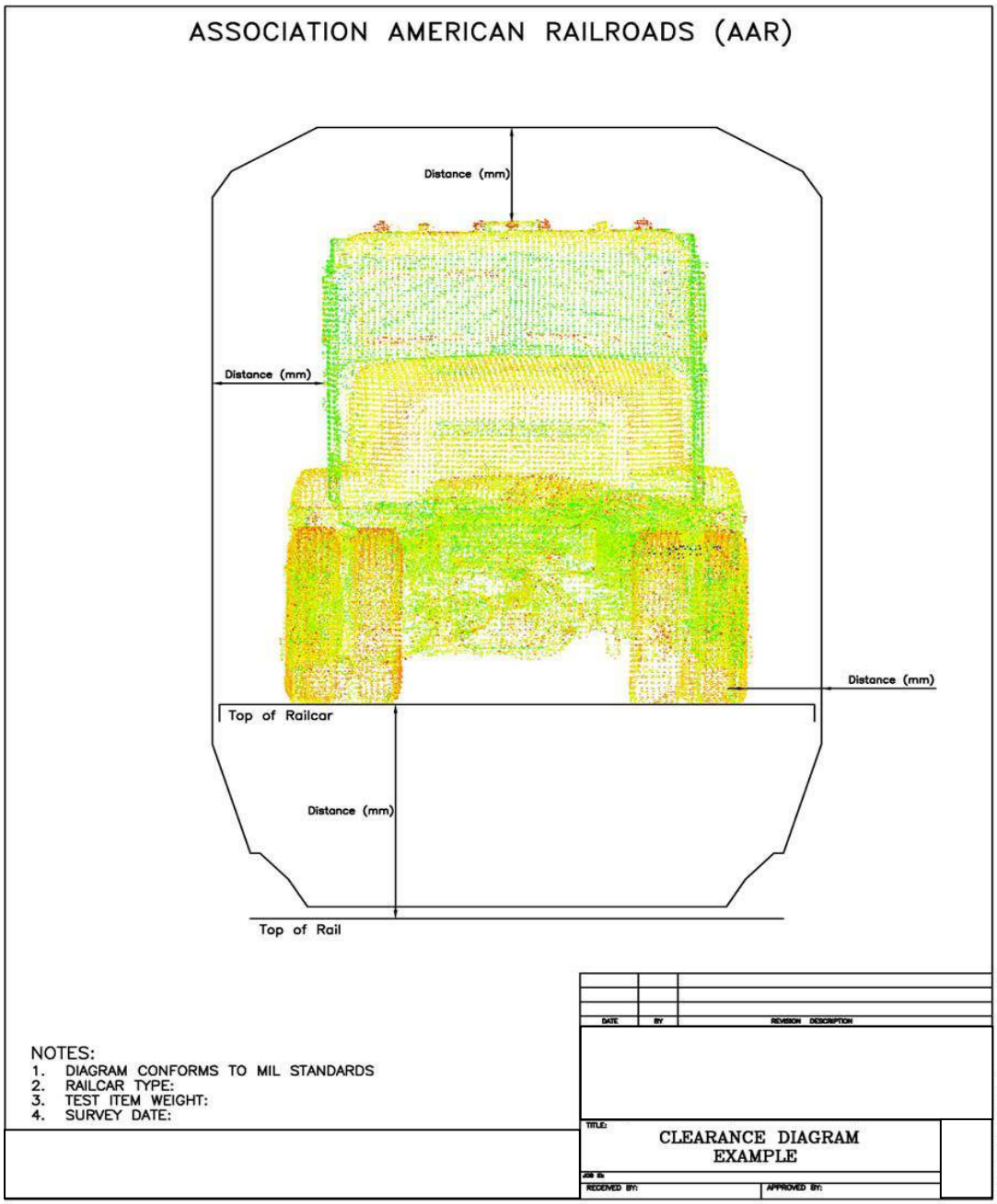


Figure 4. Representation of test item profile with applicable rail clearance diagram.

b. If required, prepare a graph of peak coupler force versus actual impact speed and compile tables of peak acceleration levels and/or plots of time versus acceleration for each impact. Present video documentation of the actual rail impacts as required for Association of American Railroads (AAR) acceptance of DOD rail impact testing. If necessary, prepare any pertinent graphs to represent other transducer measurements (i.e., strain versus time, temperature recordings, etc.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

6.3 Highway Transportability.

Summarize physical parameter measurements in tables. Describe the details of test incidents in narrative form and illustrate physical events with photographs. Verify that loading the test item does not exceed the gross vehicle and component ratings of the carrier. If feasible, compare physical characteristics with federal highway restrictions.

6.4 Marine Transportability.

Summarize and tabulate the data to present peak or critical measurements, displacements, and interferences. Include supporting photographs, sketches and curves as appropriate. Describe the details of test incidents in narrative form and illustrate physical events with photographs.

6.5 Terminals Handling and Movement.

Review and compare measured and observed data to ensure that test objectives and system criteria were achieved. Analyze the degradation of performance from pretest levels, in the case of sheltered systems or packaged equipment, or any item damage with respect to the effect on materiel utility and provisions of the applicable criteria. Analyze shock and vibration data to determine peak values and frequency content. Describe the details of test incidents in narrative form and illustrate physical events with photographs. A sample format for collecting data is provided in Table 11.

TABLE 11. SAMPLE TABLE OF ACHIEVED RIGGING PARAMETERS

Parameter	Value
Weight, lb	
Front sling length, ft	
Rear sling length, ft	
Longitudinal inclination of test item, deg	
Lateral inclination of test item, deg	
Apex height, ft	
Left front sling angle relative to ground, deg	
Right front sling angle relative to ground, deg	
Right rear sling angle relative to ground, deg	
Left rear sling angle relative to ground, deg	

6.6 Air Transportability - Fixed Wing Internal.

Analyze the measured data in accordance with MIL-STD-1791 to determine the suitability of the test item for transport on USAF and CRAF aircraft. If a test loading is conducted, describe the measured interferences between the test item and the aircraft. Once all possible interferences are resolved, the test item will be certified for air transport in a letter from the USAF ATTLA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

6.7 Air Transportability - Rotary Wing Internal.

Describe the actions taken to service and prepare the item for aircraft loading. Describe the method employed in loading and unloading the item into the aircraft. Photographs of the item during and upon completion of the loading will document the loading and tie-down configuration. Present in narrative form any incidents or interferences noted during the loading and unloading.

6.8 Air Transportability - Rotary Wing External.

Establish static characteristics for system flight as suspended in a controlled environment to simulate flight characteristics. Describe the actions taken to prepare the load for the flight demonstration in a format comparable to that used in relevant field and technical manuals. Photographs of the test item will document the rigging configuration and sling orientation. Summarize the results of the MSFDCS in tabular or narrative form, supplemented by any observations made by a review of the flight test videotape(s). Compare the results of the flight tests to the criteria listed in MIL- STD-913A. Provide the original MSFDCS to the DEVCOM SC project officer for the test file. Provide a recommendation as to the maximum safe speed for external air transport of each load configuration and helicopter combination. Identify any restrictions of external air transport load configurations if flight test results were unsatisfactory. Sample formats for collecting data are provided in Tables 12 through 14.

TABLE 12. SAMPLE TABLE OF ROTARY WING EXTERNAL SLING CONFIGURATIONS

Item Configuration	Weight, lb	Sling Set Capacity, lb	Link Count, Front/Rear	Chain Extensions Front/Rear	Single or Dual Point	Flight Direction	Maximum Airspeed, knots

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 13. SAMPLE TABLE OF HELICOPTER SLING LOAD STATIC LIFT DATA

HSL Configuration, lb	Sling Set Capacity, lb	Rigging Configuration	Link Count		Chain Extension		Flight Attitude, deg	Direction of Travel [a]	Dual-point Weight Distribution			
									Weight, lb		Percent.	
			Front	Rear	Front	Rear			Front	Rear	Front	Rear
[test item]												
	10,000	Single-point							NA			
	10,000	CH-47 Dual-point										
	10,000	CH-53 Dual-point										
[test item]												
	25,000	Single-point							NA			
	25,000	CH-47 Dual-point										
	25,000	CH-53 Dual-point										

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE 14. SAMPLE TABLE OF RIGGING PROCEDURES

Preparation. Prepare the load using the following steps:
(1) Ensure the fuel tank is not over 3/4 full. Inspect the fuel tank cap, oil filler cap, and battery caps, for proper installation.
(2) Engage the vehicle parking brake and put the transmission in neutral.
(3) Ensure the front wheels are pointed straight ahead. Tie down the steering wheel with Type III Nylon cord to the securing device provided under the dashboard.
(4) Fold mirrors to their forward position. Secure in place with Type III nylon cord tied from the support arm to the nearest hood lift provision.
(5) Secure all doors with Type III nylon cord and/or tape.
(6) Secure all cargo lashings with Type III nylon cord or tape. Secure all loose items with Type III nylon cord and/or CGU-1/B.
(7) Cover all lights, markers, and reflectors with tape.
(8) Tape the windshield in an X formation from corner to corner. [Unarmored windshield only]
Rigging. Rig the load according to the following steps:
(1) Position the apex fitting on top of the [test item]. Route the outer sling legs (1 and 2) to the front of the [test item] and inner sling legs (3 and 4) to the rear. Sling legs 1 and 3 must be on the left side of the load.
(2) Route the chain end of sling leg 1 through the front left provision. Place link [number] into the grab hook. Repeat with sling leg 2 through the front right provision.
(3) Route the chain end of sling leg 3 through the rear left provision. Place link [number] into the grab hook. Repeat with sling leg 4 through the rear right provision.
(4) Secure excess chain with tape or Type III nylon cord.
(5) Cluster and tie (breakaway technique) all sling legs together on top of the [test item] to prevent entanglement during hookup and lift-off.
Hookup
(1) The hookup team stands on top of the [test item]
(2) The static wand person discharges the static electricity with the static wand. The hookup person places the apex fitting onto the aircraft cargo hook.
(3) The hookup team dismounts the [test item], to the side.
(4) The hookup team clears the immediate vicinity of the load as the helicopter removes slack from the sling legs. When successful hookup is assured, the hookup team quickly exits the area to the designated rendezvous point.
Derigging.
Derigging is the reverse of the preparation and rigging procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

6.9 Air Transportability - Airdropped Materiel.

Summarize physical parameter measurements in a table. Identify any incompatibilities between the airdrop platform and designated aircraft. Describe the actions taken to prepare and rig the load. Photographs of the test item will document the rigging configuration. Summarize the results of each test phase in tabular or narrative form, supplemented by observations made by a review of film and videotapes. Describe the results of visual inspections and functional checks accomplished following each test phase, illustrating physical events with photographs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX A. GENERAL PHYSICAL CHARACTERISTICS FOR TRANSPORTABILITY.

TABLE A-1. GENERAL PHYSICAL CHARACTERISTICS

MEASUREMENT (DIMENSIONS)	UNIT*	TRANSPORT MODE							
		RAIL	HIGHWAY	MARINE	TERMINALS	AIR , FIXED,INTERNAL	AIR, ROTARY, INTERNAL	AIR, ROTARY, EXTERNAL	AIRDROPPED MATERIEL
Dimensions									
Length	inches	X	X	X	X	X	X	X	X
Width, overall	inches	X	X	X	X	X	X	X	X
Width, reduced	inches	X	X	X		X	X	X	X
Width, outside wheels	inches	X	X	X		X	X	X	X
Height, overall maximum	inches	X	X	X	X	X	X	X	X
Height, reduced	inches	X	X	X		X	X	X	X
Height at Front of Vehicle/Trailer	inches	X	X	X		X	X		X
Height at Rear of Vehicle/Trailer	inches	X	X	X		X	X		X
Forward Overhang [a]	inches	X	X	X		X	X		X
Forward Projection [b]	inches	X	X	X		X	X		X
Rear Overhang [c]	inches	X	X	X		X	X		X
Rear Projection [d]	inches	X	X	X		X	X		X
Forward Ground Clearance	inches	X	X	X		X	X		X
Mid-Wheelbase Ground Clearance	inches	X	X	X		X	X		X
Rear Ground Clearance	inches	X	X	X		X	X		X
Fifth Wheel/Pintle Distance [e]	inches	X	X	X		X	X		X
Fifth Wheel/Pintle Height [f]	inches	X	X	X		X	X		X
Landing Gear Pad Size (Trailers)	square inches	X	X	X		X	X		X
Kingpin/Lunette Height (Trailers)	inches	X	X	X		X	X		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX A. GENERAL PHYSICAL CHARACTERISTICS FOR TRANSPORTABILITY.

TABLE A-1. CONTINUED

MEASUREMENT	UNIT*	TRANSPORT MODE							
		RAIL	HIGHWAY	MARINE	TERMINALS	AIR , FIXED, INTERNAL	AIR, ROTARY, INTERNAL	AIR, ROTARY, EXTERNAL	AIRDROPPED MATERIEL
Wheel/Axle Measurements									
Bogie/Walking Beam Articulation [g]	degrees	X	X	X		X	X		X
Number of Axles	N/A	X	X	X		X	X		X
Distance from Front Axle to 2d Axle	inches	X	X	X		X	X		X
Distance from 2d Axle to 3d Axle	inches	X	X	X		X	X		X
Distance from 3d Axle to 4th Axle	inches	X	X	X		X	X		X
Lunette/Kingpin to first axle distance	inches	X	X	X		X	X		X
Landing gear distance	inches	X	X	X		X	X		X
Wheelbase	inches	X	X	X		X	X	X	X
Track width	inches	X	X	X		X	X	X	X
Number of Wheels on Each Axle	N/A	X	X	X		X	X		X
Tire Size	xx R zz	X	X	X		X	X		X
Tire Radius	inches	X	X	X		X	X		X
Tire Pressure	psig	X	X	X		X	X		X
Tire Contact Length	inches	X	X	X		X	X		X
Tire Contact Width	inches	X	X	X		X	X		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX A. GENERAL PHYSICAL CHARACTERISTICS FOR TRANSPORTABILITY.

TABLE A-1. CONTINUED

MEASUREMENT	UNIT*	TRANSPORT MODE							
		RAIL	HIGHWAY	MARINE	TERMINALS	AIR , FIXED, INTERNAL	AIR, ROTARY, INTERNAL	AIR, ROTARY, EXTERNAL	AIRDROPPED MATERIEL
Weight Measurements									
Curb Item Weight	pounds	X	X	X	X	X	X	X	X
Gross Item Weight	pounds	X	X	X	X	X	X	X	X
Weight of Item on Each Axle	pounds	X	X	X		X	X		X
Ratings									
Gross Item Weight Rating	pounds	X	X	X		X	X		X
Rating of Each Axle	pounds	X	X	X		X	X		X
Rating of Suspension of Each Axle	pounds	X	X	X		X	X		X
Weight Capacity of Each Transport Provision	pounds	X	X	X		X	X	X	X
Tire Load Rating at 55 MPH/ 88 KPH	pounds	X	X	X		X	X		X
Fifth Wheel/Pintle Rating	pounds	X	X	X		X	X		X
Landing Gear Rating (Trailers) (One Leg)	pounds	X	X	X		X	X		X

[a] Distance from the forward-most portion of the item at the bottom to the center of the front axle.
[b] Distance from the forward-most portion of the item at the top to the center of the front pivot point (axle, kingpin, etc.).
[c] Distance from the rearward-most portion of the item at the bottom to the center of the rear axle.
[d] Distance from the rearward-most portion of the item at the top to the center of the rear axle.
[e] Distance from the center of the front axle to the center of the fifth wheel or pintle.
[f] Height of the top of the fifth wheel or the center portion of the pintle from the ground.
[g] Maximum articulation (without wheels leaving the ground) of any multi-axled trailer or truck rear. Defines the maximum ramp angle the item can negotiate while maintaining full load distribution on all axles.

* Note: Data are required in English units for direct comparison with aircraft specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX B. FIXED WING AIR TRANSPORTABILITY REQUIREMENTS.

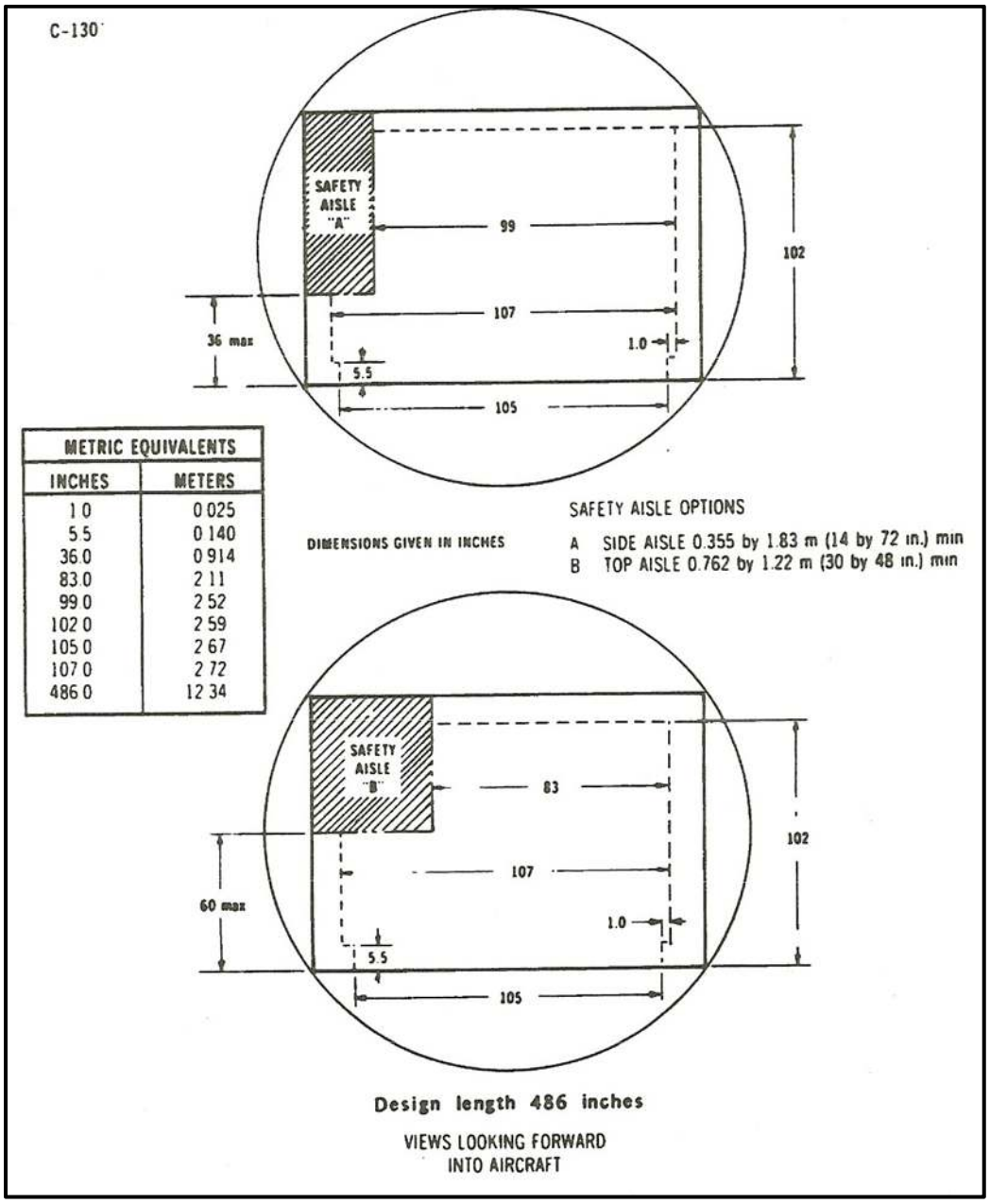


Figure B-1. C-130 cargo dimensional limits.

Note: For vehicle with tracks and non-sensitive steering, 6-in. of rolling shoring is required when the clearance between the side rails and the vehicle is critical. Vehicles with clearance of less than 2-1/2-inches on each side require ATTLA review and approval. These data are derived from MIL-STD-1791.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX B. FIXED WING AIR TRANSPORTABILITY REQUIREMENTS.

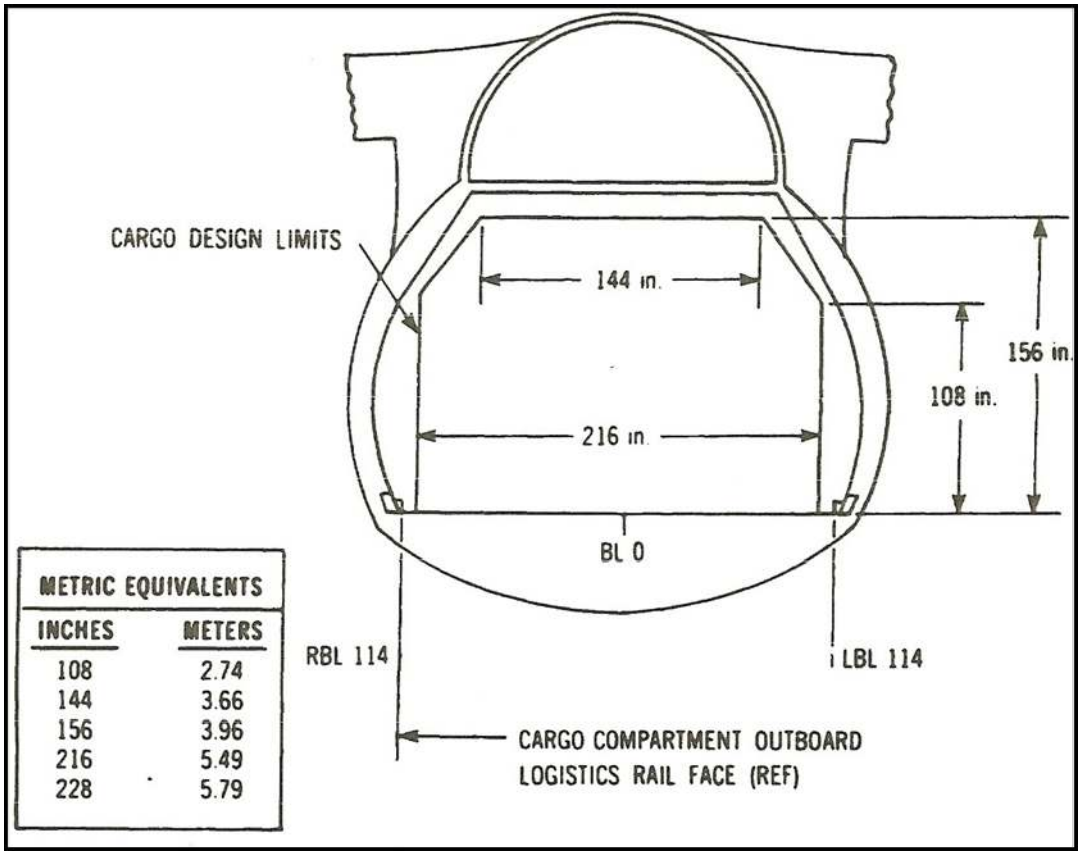


Figure B-2. C-5A/B cargo dimensional limits.

Notes: Cargo dimensional design limits shown above are the maximum deployable dimensions to be used in the design of cargo items to be transported in the C-5/B. Design length is 121 ft and 2 in. These data are derived from MIL-STD-1791.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX B. FIXED WING AIR TRANSPORTABILITY REQUIREMENTS.

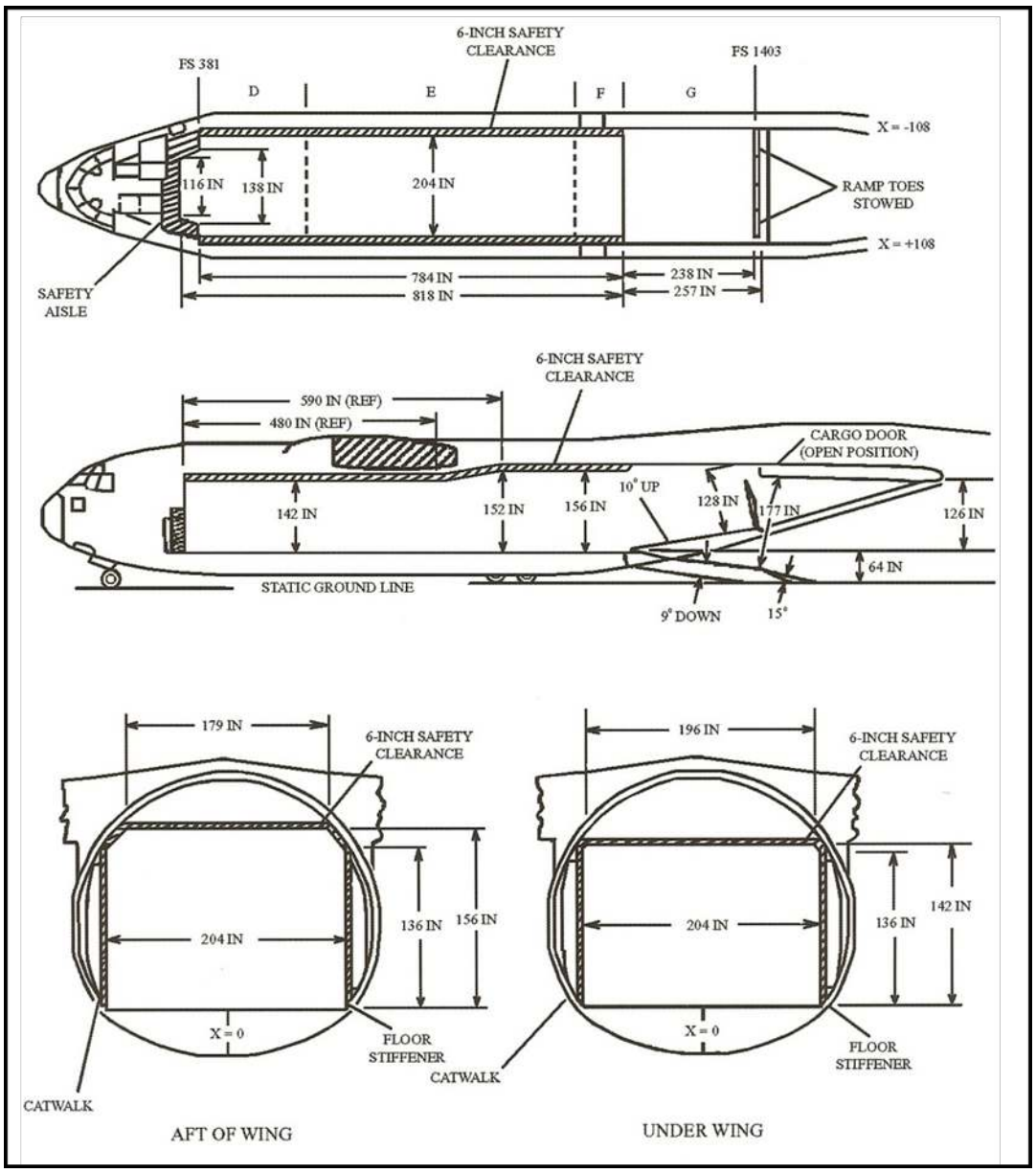


Figure B-3. C-17 cargo dimensional design limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX B. FIXED WING AIR TRANSPORTABILITY REQUIREMENTS.

TABLE B-1. FIXED WING AIRCRAFT CARGO CAPABILITY

AIRCRAFT	EFFECTIVE ACL[a]		MAXIMUM ACL[b]	
	kg	lb	kg	lb
C-130	6,800	15,000	19,050	42,000
C-5	80,740	178,000	120,200	265,000
C-17	58,970	130,000	77,110	170,000

[a] Effective allowable cabin load (ACL) is defined as the longest leg over a water route. The range and mission payload capability of the aircraft varies depending on crew, fuel and oils on board, temperature, and winds. The weight and balance (center of gravity) of the aircraft is vitally important to its stability.

[b] Maximum ACL is a contingency capability which is defined as a short-range or war-time mission. Designers should use the values for the effective ACL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX C. ROTARY-WING AIRCRAFT INTERNAL PAYLOAD CAPABILITIES.

TABLE C-1. MAXIMUM HELICOPTER INTERNAL LOAD

MISSION SCENARIO	CH-46E	CH-47D	CH-53D	CH-53E	MV-22
	lb				
Sea Level, 60F, 30 nautical miles	6,000	23,300	14,770	34,990	13,850
2,000 ft, 70F, 30 nautical miles	5,600	23,350	13,970	28,600	9,840
4,000 ft, 95F, 30 nautical miles	3,890	16,900	7,910	18,600	8,010

TABLE C-2. HELICOPTER INTERNAL TRANSPORT CONSTRAINTS[a]

HELICOPTER	HEIGHT	WIDTH	LENGTH
	in.	in.	in.
CH-46	69	72	290
CH-47D	72	80	331
CH-53E	77	90	450
MV-22	60	66	250

[a] These design limits allow for a 6-in. safety clearance between the equipment and the aircraft overhead and a 5-in. clearance between the equipment and the aircraft sidewalls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX C. ROTARY-WING AIRCRAFT INTERNAL PAYLOAD CAPABILITIES.

TABLE C-3. CONSTRAINTS FOR NAVY MARINE CORPS ROTARY-WING AIRCRAFT

	CARGO OPENING DIMENSIONS[a]			CARGO COMPARTMENT				CRASH LOAD RESTRAINT CRITERIA				
AIRCRAFT	LOCATION	WIDTH, in.	HEIGHT, in.	LENGTH, in.	WIDTH, in.	HEIGHT (AT FLOOR LINE), in.	STRUCTURAL[b] FLOOR LOAD LIMITS, lb/ft^2	FORWARD G's	AFT G's	LATERAL G's	UP G's	DOWN G's
SH-3G	Side	68	60	231	76	71	300	8	4	4	4	4
H-46A/D/E	Rear ramp	70	69	290	72	69	300	10	7.5	3	3	10
SH-608	Side	44	54	130	73	54	225	8	4	4	4	8
SH-60F	Side	44	54	130	73	54	75 Fwd cabin 300 Aft cabin	8	4	4	4	8
HH-60H	Rear ramp	44	54	130	73	54	75 Fwd cabin 300 Aft cabin	8	4	4	4	8
CH-53A	Rear ramp	96	77	450	90	77	300	10	7.5	3	3	10
CH-53D	Rear ramp	96	77	450	90	77	300	10	7.5	3	3	10
RH-53D	Rear ramp	96	77	450	90	77	300	10	7.5	3	3	10
CH-53E	Rear ramp	96	77	450	90	77[c]	300	10	7.5	3	3	10
MH-53E	Rear ramp	96	77	450	90	77[d]	300	10	7.5	3	3	10
V-22	Rear ramp	72	90	290	71	68	300	16	5	10	5	16

[a] Dimensional limits given are airframe dimensions. Clearances must be provided as follows: 6 in. at the top, 10 in. (minimum) on one side for personnel passageway, and 5 in. on the other side.

[b] Structural floor load limits for entire cargo compartment. For wheeled vehicle, conveyor, etc. load limits, see respective cargo, loading manual, Naval Air Training and Operating Procedures Standardization (NATOPS) Program Flight Manual, Aircraft Weight and Balance Manual for each aircraft.

[c] 73 in. when single point cargo hook is stowed and 75 in. unstowed.

[d] 58 in. aft cabin when Mine Counter Measure (MCM) equipment is installed.

Note: This table was derived from MIL-STD-1366E.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

TABLE C-4. CONSTRAINTS FOR ARMY ROTARY-WING AIRCRAFT

AIRCRAFT	CARGO OPENING DIMENSIONS[a]			CARGO COMPARTMENT[c]				CRASH LOAD RESTRAINT CRITERIA				
	LOCATION	WIDTH, in.	HEIGHT, in.	LENGTH, in.	WIDTH, in.	HEIGHT, in.	STRUCTURAL FLOOR LOAD LIMITS, lb/ft^2	FORWARD G's	AFT G's	LATERAL G's	UP G's	DOWN G's
UH-1H	Each side	92/74[b]	49	82/47.5[d]	77/15[d]	46	100	8	8	8	4	8
UH-60 (A/L/K)	Each side	68	52	92/127.5[e]	50/46[e]	46	300	12	3	8	3	-
CH-47 (C/D/E)	Rear ramp	90	78	331	80	72	300 (or 2500 on treadway)	4	2	1.5	2	4

[a] Dimensions given are airframe dimensions. Additional clearance of 6 in. between the equipment and the aircraft ceiling must be provided.

[b] Dimensions are for both doors and sliding door opened respectively.

[c] Dimensions allow a 5 in. clearance inside the centerline of the perimeter tie-down fittings and clearance of 6 in. between the equipment and the aircraft ceiling.

[d] The first length and width dimensions are for the total cargo compartment. However, they do not allow for transmission box. The second length dimension is the unobstructed allowable cargo length form front of cargo compartment to transmission box. The second width dimension is the allowable cargo width for the areas on either side of the transmission box.

[e] Second dimension includes gunner's area.

Note: This table was derived from MIL-STD-1366E.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX D. ROTARY-WING AIRCRAFT EXTERNAL PAYLOAD CAPABILITIES.

TABLE D-1. MAXIMUM HELICOPTER EXTERNAL LOAD

MISSION SCENARIO	UH-1H	UH-60A	UH-60L	CH-46E	CH-47D	CH-53D	CH-53E	MV-22[a]
	lb	lb	lb	lb	lb	lb	lb	lb
Sea Level, 60F, 30 nautical miles	2,585	7,843	9,000	5,915	23,324	14,700	34,770	13,320
2,000 ft, 70F, 30 nautical miles	2,624	7,302	9,000	5,480	23,396	13,900	28,300	9,330
4,000 ft, 95F, 30 nautical miles	1,169	4,700	6,630	3,780	16,644	7,860	18,200	7,500

[a] The Marine V-22 helicopter is under-development.

TABLE D-2. HELICOPTER CARGO HOOK LOAD RATINGS[a]

HELICOPTER	HOOK(S) LOAD RATINGS, LB
UH-1H	4,000
UH-60A/L	9,000
CH-46E	10,000
CH-47D	26,000 (center hook)/ 17,000 (fore and aft hooks)
CH-53E	36,000
MV-22	15,000 (single hook)/ 10,000 (dual hook)

[a] Although the primary limitation to external air transport is the available payload (Table C-1), an additional limitation is the helicopter hook strength.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.

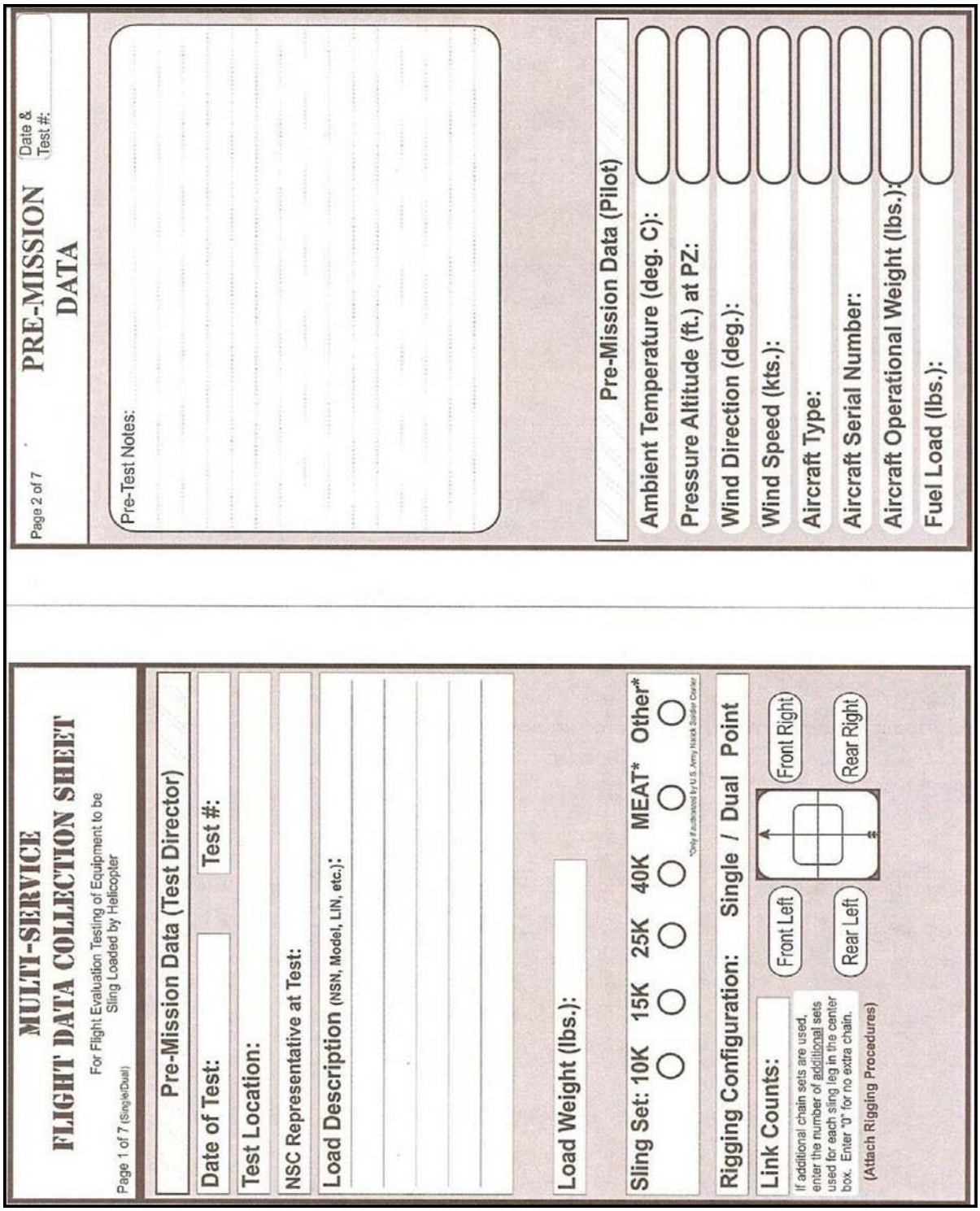


Figure E-1. MSFDCS and rating criteria.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.

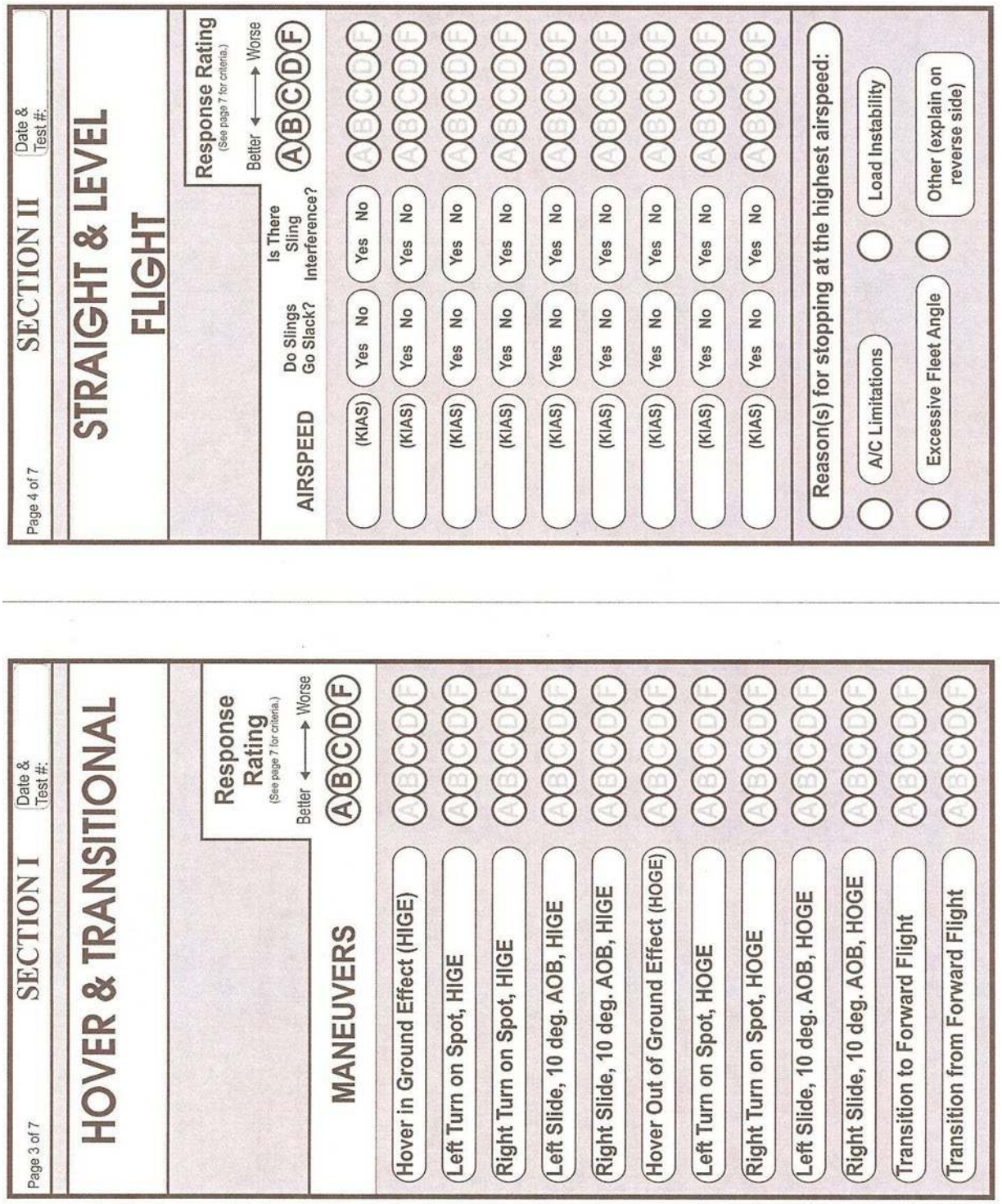
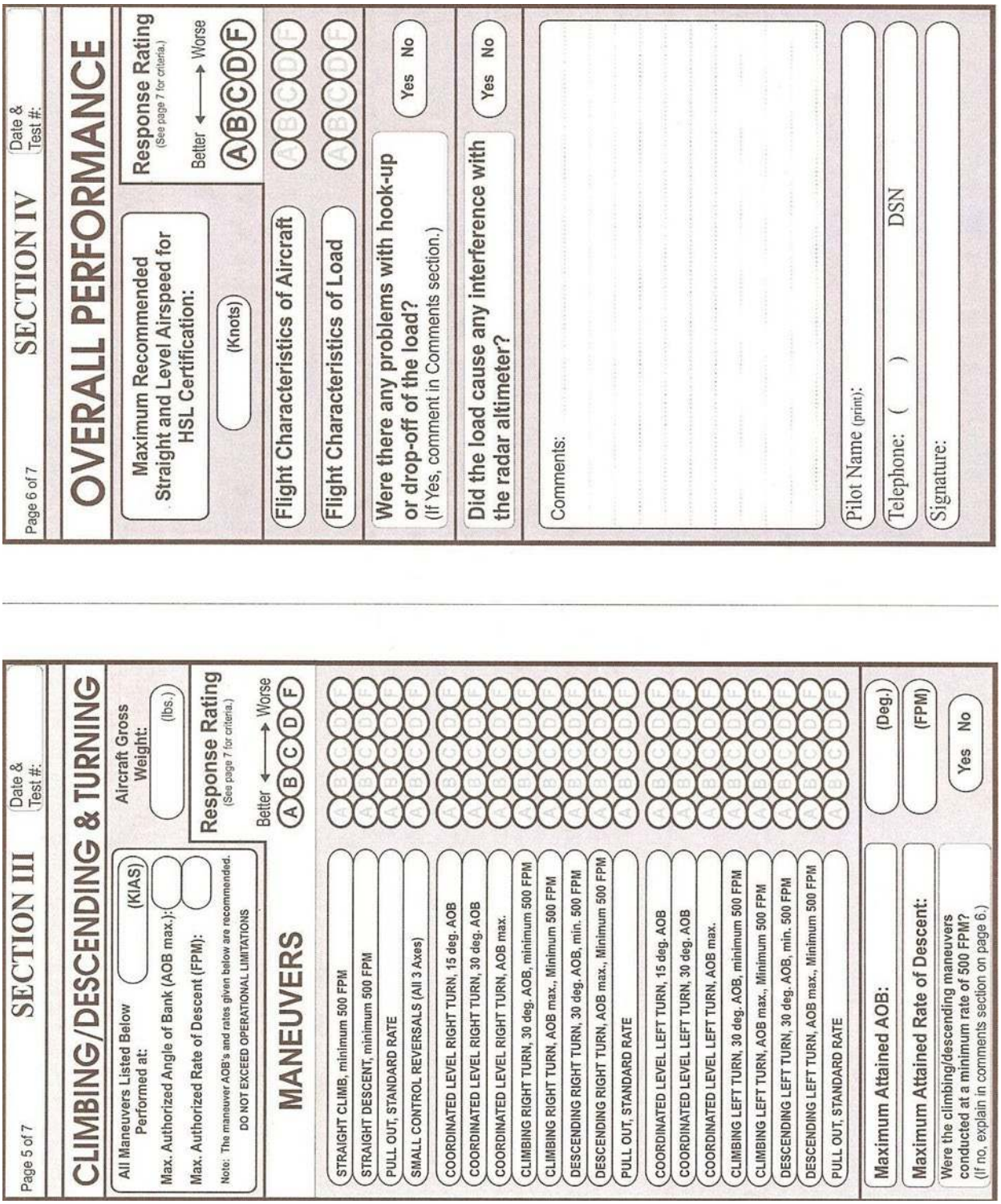


Figure E-1. Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.



Page 5 of 7 SECTION III Date & Test #:

CLIMBING/DESCENDING & TURNING

All Maneuvers Listed Below Performed at: (KIAS)

Max. Authorized Angle of Bank (AOB max.):

Max. Authorized Rate of Descent (FPM):

Note: The maneuver AOB's and rates given below are recommended. DO NOT EXCEED OPERATIONAL LIMITATIONS

Aircraft Gross Weight: (lbs.)

Response Rating (See page 7 for criteria.)

Better ← → Worse

MANEUVERS	A	B	C	D	F
STRAIGHT CLIMB, minimum 500 FPM	○	○	○	○	○
STRAIGHT DESCENT, minimum 500 FPM	○	○	○	○	○
PULL OUT, STANDARD RATE	○	○	○	○	○
SMALL CONTROL REVERSALS (All 3 Axes)	○	○	○	○	○
COORDINATED LEVEL RIGHT TURN, 15 deg. AOB	○	○	○	○	○
COORDINATED LEVEL RIGHT TURN, 30 deg. AOB	○	○	○	○	○
COORDINATED LEVEL RIGHT TURN, AOB max.	○	○	○	○	○
CLIMBING RIGHT TURN, 30 deg. AOB, minimum 500 FPM	○	○	○	○	○
CLIMBING RIGHT TURN, AOB max., Minimum 500 FPM	○	○	○	○	○
DESCENDING RIGHT TURN, 30 deg. AOB, min. 500 FPM	○	○	○	○	○
DESCENDING RIGHT TURN, AOB max., Minimum 500 FPM	○	○	○	○	○
PULL OUT, STANDARD RATE	○	○	○	○	○
COORDINATED LEVEL LEFT TURN, 15 deg. AOB	○	○	○	○	○
COORDINATED LEVEL LEFT TURN, 30 deg. AOB	○	○	○	○	○
COORDINATED LEVEL LEFT TURN, AOB max.	○	○	○	○	○
CLIMBING LEFT TURN, 30 deg. AOB, minimum 500 FPM	○	○	○	○	○
CLIMBING LEFT TURN, AOB max., Minimum 500 FPM	○	○	○	○	○
DESCENDING LEFT TURN, 30 deg. AOB, min. 500 FPM	○	○	○	○	○
DESCENDING LEFT TURN, AOB max., Minimum 500 FPM	○	○	○	○	○
PULL OUT, STANDARD RATE	○	○	○	○	○

Maximum Attained AOB: (Deg.)

Maximum Attained Rate of Descent: (FPM)

Were the climbing/descending maneuvers conducted at a minimum rate of 500 FPM? (If no, explain in comments section on page 6.) Yes No

Page 6 of 7 SECTION IV Date & Test #:

OVERALL PERFORMANCE

Maximum Recommended Straight and Level Airspeed for HSL Certification: (Knots)

Response Rating (See page 7 for criteria.)

Better ← → Worse

	A	B	C	D	F
Flight Characteristics of Aircraft	○	○	○	○	○
Flight Characteristics of Load	○	○	○	○	○

Were there any problems with hook-up or drop-off of the load? (If Yes, comment in Comments section.) Yes No

Did the load cause any interference with the radar altimeter? Yes No

Comments:

Pilot Name (print):

Telephone: ( ) DSN

Signature:

Figure E-1. Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.

Page 7 of 7

RATING CRITERIA FOR AIRCRAFT FLIGHT CHARACTERISTICS WITH EXTERNAL LOADS

A. Excellent handling qualities. Effects of the load upon the aircraft performance are negligible at the prescribed airspeed.

B. Good handling qualities. Effects of the load on the aircraft performance are noticeable, but require little or no effort from the pilot to maintain control of the aircraft at the prescribed airspeed.

C. Fair handling qualities. Effects of the load on the aircraft performance are moderate, but readily controllable. The pilot should exercise moderate caution and pay close attention to the effects of the load on the aircraft in order to maintain control of the aircraft at the prescribed airspeed.

D. Poor handling qualities. Effects of the load on the aircraft performance are significant and require constant attention from the pilot to control the aircraft. Caution must be maintained at all times in order to control the aircraft at the prescribed airspeed.

F. Unacceptable handling qualities. Flight under these conditions is dangerous and requires constant attention from the pilot to avoid loss of control of the aircraft. Aircraft is constantly unstable. Flight at this or higher air speed is not recommended.

RATING CRITERIA FOR EXTERNAL LOAD STABILITY CHARACTERISTICS DURING FLIGHT

A. Excellent Load Stability: Load maintains directional stability throughout maneuvers. Minimal load oscillation and/or minimal load rotation or weathervaning. Requires minimal concentration by the flight crew.

B. Good Load Stability: Load maintains directional stability for most maneuvers. Only moderate load oscillation and/or moderate load rotation or weathervaning occurs. Requires minimal concentration by the flight crew.

C. Fair Load Stability: Load may oscillate, rotate and/or weathervane during most maneuvers. Directional orientation is not stable throughout maneuvers. However the load remains stable in its rotational state, the rotation does not continue to wind up the sling legs, and the load does not pose a threat to the aircraft.

D. Poor Load Stability: Load oscillates, rotates, or weathervanes during all maneuvers. Directional instability may become severe and require immediate action by the flight crew to prevent damage to the load and/or aircraft, or danger to personnel.

F. Unacceptable Load Stability: Load is uncontrollable for most or all of the maneuvers. Directional instability is unpredictable and dangerous. Transport of the load at the prescribed airspeed is not recommended.

Figure E-1. Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.

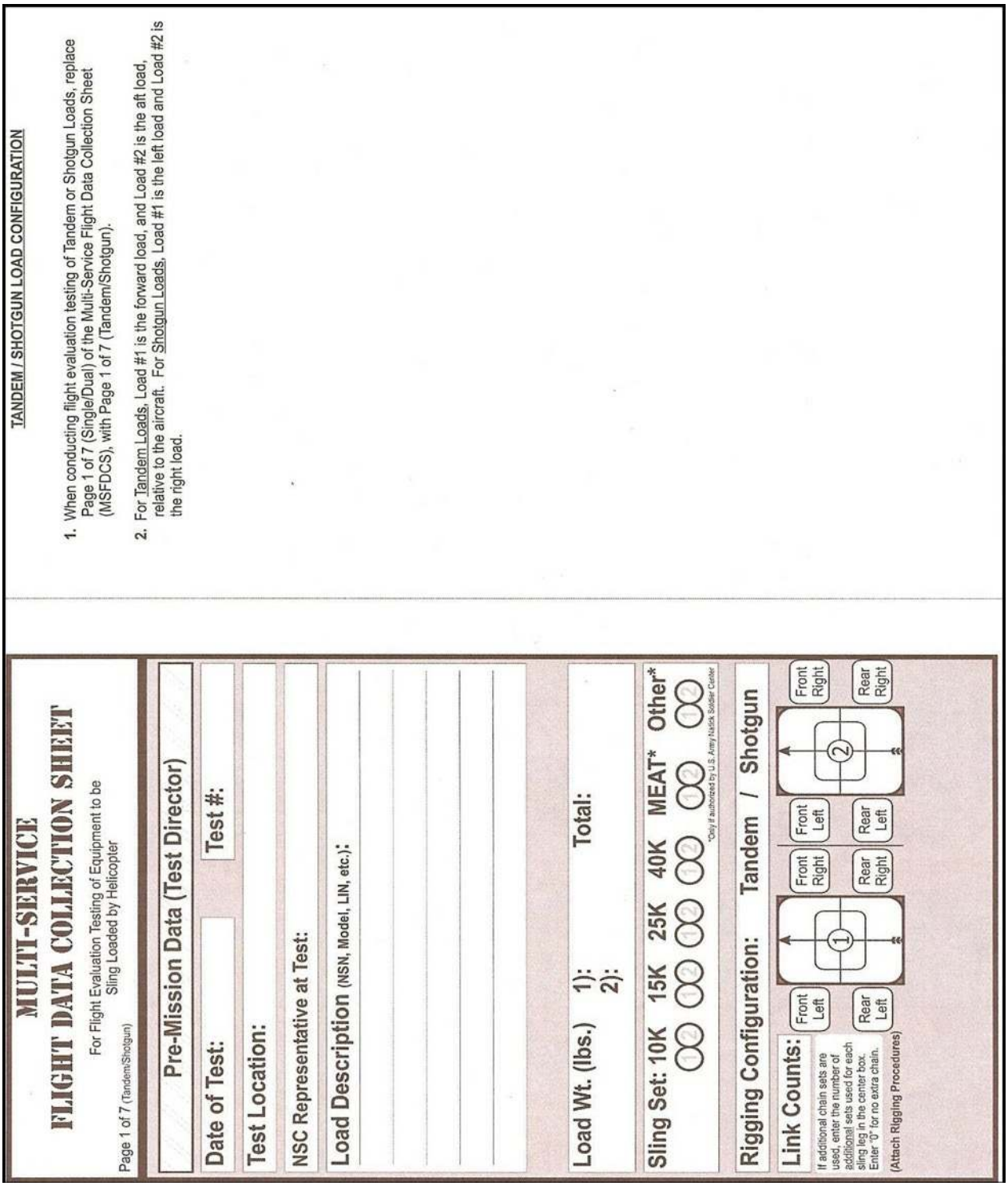


Figure E-1. Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX E. EXTERNAL AIR TRANSPORTABILITY MULTI-SERVICE FLIGHT DATA COLLECTION SHEET (MSFDCS) AND RATING CRITERIA.

TABLE E-1. HSL SUPPORTING DOCUMENTS

DOCUMENT	TIME FRAME
Draft DTP, DEVCOM SC Concurrence Form, RSR, and any other supporting information (e.g., MSFDCS from previous HSL test) delivered to RTC and AEC.	Test - 6 weeks
Final DTP, DEVCOM SC Concurrence Form, RSR, and any other supporting information delivered to RTC and AEC.	Test - 4 weeks
RTC delivers correspondence documentation (concur/non-concur with ATC RSR) to AEC and ATC.	Test - 2 weeks
AEC delivers SR to ATC.	Prior to test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX F. ABBREVIATIONS.

AAR	Association of American Railroads
ACC	Air Combat Command
ACL	allowable cabin load
AEC	U.S. Army Evaluation Center
AFB	Air Force Base
AFSOC	Air Force Special Operations Command
AMC	Air Mobility Command
AMCP	Air Mobility Command Publication
ANSI	American National Standards Institute
AR	Army Regulation
ASC	Aeronautical Systems Center
ASFS	Army Standard Family of Shelters
ASTM	American Society of Testing and Materials
ATC	U.S. Army Aberdeen Test Center
ATTLA	Air Force Transportability Test Loading Activity
BII	basic issue items
°C	degrees Celsius
CFR	Code of Federal Regulations
CG	center of gravity
CLA	Cargo Loading Adaptability
cm	centimeter
CRAF	Civil Reserve Air Fleet
DEVCOM	Development and Engineering Center
DOD	Department of Defense
DODI	Department of Defense Instruction
DOT	Department of Transportation
DTS	Defense Transportation System
EAT	external air transport
°F	degrees Fahrenheit
ft	feet
ft/sec	feet per second
FM	Field Manual
FMVSS	Federal Motor Vehicle Safety Standard
FR	France
GAWR	gross axle weight rating
GCWR	gross combination weight rating
GE	Germany

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX F. ABBREVIATIONS.

GSW	gross shipping weight
GVWR	gross vehicle weight rating
GWR	gross weight rating
HSL	helicopter sling load
IAT	internal air transport
in.	inch
ISO	International Organization for Standardization
ITOP	International Test Operations Procedure
JAI	Joint Aircraft Inspection
kg	kilogram
km/hr	kilometers per hour
kN	kilonewton
kPa	kilopascal
LAPE	Low Altitude Parachute Extraction
LASH	Lighter Aboard Ship
lb	pound
lbf	pounds force
LOLO	lift on/lift off
LOTS	Logistics-Over-the-Shore
LVAD	Low Velocity Air Drop
m	meter
m/sec	meters per second
MCM	Mine Counter Measure
MHE	material handling equipment
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
mph	miles per hour
MSFDCS	Multi-Service Flight Data Collection Sheet
NATO	North Atlantic Treaty Organization
NATOPS	Naval Air Training and Operating Procedures Standardization
NAVAIR	Naval Air Systems Command
NHTSA	National Highway Traffic Safety Administration
OVE	On-Vehicle Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX F. ABBREVIATIONS.

psi	pounds per square inch
psig	pounds per square inh gauge
PTP	Proposed Test Plan
RORO	roll on/roll off
RTC	U.S. Army Redstone Test Center
SAIT	Simulated Airdrop Impact Test
SC	Soldier Center
SDDCTEA	Military Surface Deployment and Distribution Command Transportation Engineering Agency
SEABEE	Sea Barge
SOP	Standing Operating Procedure
STRACNET	Strategic Rail Corridor Network
TEA	Transportation Engineering Agency
TM	Technical Manual
TOP	Test Operations Procedure
TSARC	Test Schedule and Review Committee
UK	United Kingdom of Great Britain and Northern Ireland
U.S.	United States
USAF	United States Air Force
USMC	United States Marine Corps
USN	United States Navy
V/STOL	Vertical/Short Take-Off and Landing

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX G. REFERENCES.

1. DODI 4540.07, Operation of the DOD Engineering for Transportability and Deployability Program, Change 3, 30 October 2019.

2. MIL-STD-1366E, DOD: Interface Standard for Transportability Criteria, 31 October 2006.

3. ISO Standard 1496/1:2013, Freight Containers - Specifications and Testing, July 2013.

4. Unified Facilities Code (UFC) 4-860-03, Railroad Track Maintenance & Safety Standards, 13 February 2008.

5. RR-C-271G, Federal Specification: Chains and Attachments, Carbon and Alloy Steel, 28 March 2016.

6. MIL-STD-209K, DOD: Interface Standard for Lifting and Tie-down Provisions, 22 February 2005.

7. National Highway Traffic Safety Administration (NHTSA), Department of Transportation (DOT), Chapter 49 Code of Federal Regulations (CFR), Chapter V, Section 571.3 Definitions, October 2011.

8. MIL-STD-913A, DOD Design Criteria Standard: Requirements for the Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopters, 3 February 1997.

9. MIL-STD-814D (with Notice 1), DOD Interface Standard: Requirements for Tie-down, Suspension, and Extraction Provisions on Military Materiel for Airdrop, 11 June 2002.

10. MIL-DTL-25959H - Detail Specification: Tie Down, Tensioners, Cargo, Aircraft, 13 November 2012.

11. SDDCTEA MI 55-19, Tiedown Instructions for Rail Movements, July 2015.

12. TOP 01-2-510, Logistics-Over-the-Shore (LOTS), 2 March 1976.

13. TEA Pamphlet 55-22, Lashing Handbook for Marine Movements, July 2009.

14. ASTM E1925, 2018 Edition, Specification for Engineering and Design Criteria for Rigid Wall Relocatable Structures, 1 October 2018.

15. ASTM D6055-96, Standard Test Methods for Mechanical Handling of Unitized Loads and Large Shipping Cases and Crates, 2019.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX G. REFERENCES.

16. ASTM D6179-20, Standard Test Methods for Mechanical Handling of Unitized Loads and Large Shipping Cases and Crates, 2020.

17. France (FR)/Germany (GE)/ United Kingdom of Great Britain and Northern Ireland (UK)/United States (US) ITOP 04-2-601 (with Change Notice 1), Drop Tests for Munitions, 13 May 2004.

18. MIL-STD-810H, DOD Test Method Standard: Environmental Engineering Considerations and Laboratory Tests, 31 January 2019.

19. FR/GE/UK/U.S. ITOP 04-2-602 (with Change Notice 1), Rough Handling Tests, 27 September 2004.

20. TOP 02-2-537, Cargo Loading Adaptability (CLA), 15 April 1971.

21. ASTM D4169-16, Standard Practice for Performance Testing of Shipping Containers and Systems, 2016.

22. MIL-STD-1791C, DOD Interface Standard: Designing for Internal Aerial Delivery in Fixed-Wing Aircraft, October 2017.

23. FM 55-450-2, Army Helicopter Internal Load Operations, 5 June 1992.

24. Technical Manual (TM) 4-48.09, Marine Corps Reference Publication (MCRP) 4-11.3E Volume (Vol.) I, Navy Tactics, Techniques, and Procedures (NTTP) 3-04.11, Air Force Joint Manual (AFJMAN) 11-223 (I), Vol. I, Commandant Instruction (COMDTINST) M13482.2B, Multiservice Helicopter Sling Load: Basic Operations and Equipment, July 2012.

25. TM 10-500-2, General Information for Rigging Airdrop Platforms, November 1990.

26. MIL-HDBK-669B, DOD Handbook: Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel, November 2010.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX G. REFERENCES.

For information only (related publications).

a. Air Mobility Command Publication (AMCP) 76-2, Airlift Planning Factors, May 1987.

b. DOD Directive 5000.1, The Defense Acquisition System, 9 September 2020.

c. DOD Directive 4510.11, DOD Transportation Engineering, 1 December 2006.

d. Army Regulation (AR) 70-47, Research, Development, and Acquisition: Engineering for Transportability Program, 11 January 2019.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

APPENDIX H. APPROVAL AUTHORITY.

CSTE-CI 23 February 2022

MEMORANDUM FOR

Commander, U.S. Army Operational Test Command
Director, U.S. Army Evaluation Center
Commanders, ATEC Test Centers
Technical Directors, ATEC Test Centers

SUBJECT: Test Operations Procedure 01-2-500B Transportability

1. Test Operations Procedure (TOP) 01-2-500B Transportability, has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP provides guidance for preparing test plans and conducting test programs to evaluate the transportability characteristics of military equipment whether towed, self-propelled, or moved by carrier over highway, off-road terrain, railway, waterway, or by air.

3. This document is approved for publication and has been posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atc.army.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army Test and Evaluation Command (CSTE-CI), 6617 Aberdeen Boulevard-Third Floor, Aberdeen Proving Ground, MD 21005-5001; or e-mailed to usarmy.apg.atec.mbx.atec-standards@mail.mil.

ZWIEBEL.MICHAEL.J.1229197289 Digitally signed by ZWIEBEL.MICHAEL.J.1229197289 Date: 2022.03.01 12:48:15 -05'00'

MICHAEL J. ZWIEBEL
Director, Directorate for Capabilities Integration (DCI)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: Policy and Standardization Division (CSTE-CI-P), U.S. Army Test and Evaluation Command, 6617 Aberdeen Boulevard, Aberdeen Proving Ground, Maryland 21005-5001. Technical information may be obtained from the preparing activity: Aviation & Foreign Systems Division (TEAT-WFF), U.S. Army Aberdeen Test Center, 6943 Colleran Rd, Aberdeen Proving Ground, MD 21005-5059. Additional copies can be requested through the following website: https://www.atec.army.mil/publications/documents.html, or through the Defense Technical Information Center, 8725 John J. Kingman Rd., STE 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:46Z
Check the source to verify that this is the current version before use.